

Document 2 - FRS - Heart Rate Monitor Application

Application: Heart Rate Monitor for Android and iOS

Document state: DRAFT / uncontrolled / deliberately messy for parser testing

Note: This is synthetic test material only. It is not approved medical-device validation content.

Revision table is mixed below; some rows intentionally repeat or omit values.

Version	Date	Author	Change
0.1	2026-05-18	QA/BA	Initial messy draft
1.0?	TBD	SME?	Imported from old template
0.2	missing	AI Test	duplicate version row

This FRS converts user needs into functional behavior for heart-rate capture, display, alerts, history, exports, authentication, privacy, and device connectivity. Formatting and numbering are deliberately inconsistent.

FRS Section 2 - Cover / uncontrolled draft

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- Old requirement copied from web portal project - review if still valid.
- Old requirement copied from web portal project - review if still valid.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
FRS-002-01	Biometric login option	Cloud Sync	Not ranked	Audit/event record shall be generated where applicable.	Need SME
FRS-002-02	Audit user actions	Android	Low	User shall receive clear error message on failure.	Pending QA
FRS-002-03	Notification retry and suppression	Android/iOS	TBD	Result must be visible in mobile UI within expected time.	Draft
FRS-002-04	Store readings securely on device	Android	TBD	Result must be visible in mobile UI within expected time.	Draft

FRS Section 3 - Abbreviations mixed list

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- FRS-003-1: App shall process heart rate data and preserve timestamp/timezone context.

FRS-003-2: App shall process heart rate data and preserve timestamp/timezone context.

- Unnumbered note - check sensor disconnect behavior during active monitoring.

- FRS-003-4: App shall process heart rate data and preserve timestamp/timezone context.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
FRS_3.1	Store readings securely on device	Bluetooth Sensor	TBD	User shall receive clear error message on failure.	Rejected once
FRS-003-02	Consent capture and privacy notice	Local DB	TBD	Data shall be encrypted in transit and at rest.	Pending QA
FRS-003-03	Show trend chart for 7/30/90 days	Notification Service	TBD	User shall receive clear error message on failure.	Draft

FRS Section 4 - Document purpose

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- Old requirement copied from web portal project - review if still valid.
 - For iOS only? For Android only? unclear; parser should capture platform if visible.
 - Unnumbered note - check sensor disconnect behavior during active monitoring.
2. Acceptance: expected result to be confirmed by QA; evidence screenshot may be attached later.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
FRS-004-01	Handle offline mode	Backend API	TBD	System shall complete the action without data loss.	Approved?
FRS-004-02	Consent capture and privacy notice	Local DB	Not ranked	TBD by clinical SME - intentionally incomplete.	Rejected once
FRS-004-03	Sync readings to cloud API	Android	Not ranked	Result must be visible in mobile UI within expected time.	Pending QA
FRS-004-04	Notification retry and suppression	Notification Service	High	User shall receive clear error message on failure.	Rejected once
FRS-004-05	Biometric login option	Android/iOS	TBD	System shall complete the action without data loss.	Approved?

FRS Section 5 - Scope and non-scope

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For iOS only? For Android only? unclear; parser should capture platform if visible.

3. Unnumbered note - check sensor disconnect behavior during active monitoring.
- Unnumbered note - check sensor disconnect behavior during active monitoring.
4. Unnumbered note - check sensor disconnect behavior during active monitoring.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
FRS_5.1	Capture heart rate from paired device	Backend API	Not ranked	Audit/event record shall be generated where applicable.	Need SME
FRS_5.2	Battery and Bluetooth warning	Cloud Sync	Not ranked	Audit/event record shall be generated where applicable.	Pending QA
FRS-005-03	Biometric login option	Cloud Sync	Not ranked	Audit/event record shall be generated where applicable.	Draft

FRS Section 6 - System overview

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- Acceptance: expected result to be confirmed by QA; evidence screenshot may be attached later.

- 5. Unnumbered note - check sensor disconnect behavior during active monitoring.
- 6. FRS-006-3: App shall process heart rate data and preserve timestamp/timezone context.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
FRS-006-01	Notification retry and suppression	Cloud Sync	TBD	Result must be visible in mobile UI within expected time.	In Review
FRS_6.2	Display current BPM with timestamp	Android	Low	Data shall be encrypted in transit and at rest.	Need SME
FRS_6.3	Store readings securely on device	Cloud Sync	TBD	Audit/event record shall be generated where applicable.	In Review
FRS-006-04	Validate sensor data range	Backend API	TBD	Audit/event record shall be generated where applicable.	Need SME

FRS Section 7 - Mobile app overview

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FRS-007-1: App shall process heart rate data and preserve timestamp/timezone context.

For iOS only? For Android only? unclear; parser should capture platform if visible.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
FRS-007-01	Validate sensor data range	Android/iOS	TBD	Audit/event record shall be generated where applicable.	Draft
FRS-007-02	Export readings as PDF/CSV	Bluetooth Sensor	Not ranked	System shall complete the action without data loss.	Draft
FRS-007-03	Store readings securely on device	Android	Low	User shall receive clear error message on failure.	Pending QA

FRS Section 8 - User roles

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7. Acceptance: expected result to be confirmed by QA; evidence screenshot may be attached later.

- Unnumbered note - check sensor disconnect behavior during active monitoring.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
FRS_8.1	Store readings securely on device	Android	TBD	User shall receive clear error message on failure.	Approved?
FRS-008-02	Validate sensor data range	Android	TBD	User shall receive clear error message on failure.	Rejected once
FRS-008-03	Support caregiver view	Backend API	TBD	Result must be visible in mobile UI within expected time.	Draft
FRS-008-04	Store readings securely on device	Android	Medium	Audit/event record shall be generated where applicable.	Draft

FRS Section 9 - Regulatory notes

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- FRS-009-1: App shall process heart rate data and preserve timestamp/timezone context.
 - For iOS only? For Android only? unclear; parser should capture platform if visible.
8. FRS-009-3: App shall process heart rate data and preserve timestamp/timezone context.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
FRS-009-01	Battery and Bluetooth warning	Android	TBD	User shall receive clear error message on failure.	Approved?
FRS-009-02	Export readings as PDF/CSV	Bluetooth Sensor	High	User shall receive clear error message on failure.	Need SME
FRS-009-03	Manual heart rate entry when device unavailable	Bluetooth Sensor	Medium	Data shall be encrypted in transit and at rest.	In Review
FRS-009-04	Battery and Bluetooth warning	Backend API	TBD	User shall receive clear error message on failure.	Need SME

MIXED TRACEABILITY LINE: URS-00X -> FRS?? -> DS component maybe -> IQ/OQ/PQ evidence TBD. This line is intentionally noisy.

FRS Section 10 - Data privacy notes

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FRS-010-1: App shall process heart rate data and preserve timestamp/timezone context.

9. Acceptance: expected result to be confirmed by QA; evidence screenshot may be attached later.

10. FRS-010-3: App shall process heart rate data and preserve timestamp/timezone context.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
FRS-010-01	Capture heart rate from paired device	Bluetooth Sensor	Low	TBD by clinical SME - intentionally incomplete.	Approved?
FRS_10.2	Validate sensor data range	Notification Service	Low	Result must be visible in mobile UI within expected time.	Pending QA
FRS-010-03	Store readings securely on device	Notification Service	High	Data shall be encrypted in transit and at rest.	Approved?

FRS Section 11 - High level assumptions

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- Old requirement copied from web portal project - review if still valid.
- For iOS only? For Android only? unclear; parser should capture platform if visible.

Old requirement copied from web portal project - review if still valid.

For iOS only? For Android only? unclear; parser should capture platform if visible.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
FRS-011-01	Generate high/low heart-rate alert	Backend API	Medium	User shall receive clear error message on failure.	Pending QA
FRS-011-02	Notification retry and suppression	Notification Service	TBD	User shall receive clear error message on failure.	Rejected once
FRS-011-03	Export readings as PDF/CSV	Notification Service	Low	Data shall be encrypted in transit and at rest.	Draft
FRS-011-04	Display current BPM with timestamp	Cloud Sync	Medium	System shall complete the action without data loss.	Need SME

FRS Section 12 - Cover / uncontrolled draft

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- For iOS only? For Android only? unclear; parser should capture platform if visible.

Acceptance: expected result to be confirmed by QA; evidence screenshot may be attached later.

11. Unnumbered note - check sensor disconnect behavior during active monitoring.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
FRS-012-01	Biometric login option	Local DB	Low	Result must be visible in mobile UI within expected time.	Draft
FRS_12.2	Support caregiver view	Notification Service	Medium	System shall complete the action without data loss.	Draft
FRS-012-03	Handle offline mode	iOS	Low	Result must be visible in mobile UI within expected time.	Rejected once
FRS-012-04	Export readings as PDF/CSV	iOS	High	System shall complete the action without data loss.	Rejected once

FRS Section 13 - Abbreviations mixed list

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12. Acceptance: expected result to be confirmed by QA; evidence screenshot may be attached later.

- Acceptance: expected result to be confirmed by QA; evidence screenshot may be attached later.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
FRS-013-01	Sync readings to cloud API	iOS	Low	Result must be visible in mobile UI within expected time.	In Review
FRS-013-02	Validate sensor data range	iOS	High	Audit/event record shall be generated where applicable.	Approved?
FRS-013-03	Validate sensor data range	Backend API	Low	User shall receive clear error message on failure.	In Review

Random inserted note: duplicate title appears again below.

FRS Section 13 - Abbreviations mixed list

FRS Section 14 - Document purpose

RAG TEST CHUNK

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Old requirement copied from web portal project - review if still valid.

For iOS only? For Android only? unclear; parser should capture platform if visible.

- FRS-014-3: App shall process heart rate data and preserve timestamp/timezone context.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
FRS_14.1	Consent capture and privacy notice	Android/iOS	TBD	User shall receive clear error message on failure.	Need SME
FRS-014-02	Battery and Bluetooth warning	Notification Service	Not ranked	Data shall be encrypted in transit and at rest.	Need SME
FRS-014-03	Biometric login option	Backend API	High	Audit/event record shall be generated where applicable.	Approved?
FRS-014-04	Consent capture and privacy notice	Cloud Sync	Low	Result must be visible in mobile UI within expected time.	Pending QA
FRS-014-05	Validate sensor data range	Local DB	High	System shall complete the action without data loss.	Rejected once

FRS Section 15 - Exception handling

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For iOS only? For Android only? unclear; parser should capture platform if visible.

- For iOS only? For Android only? unclear; parser should capture platform if visible.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
FRS_15.1	Support caregiver view	Android	Not ranked	User shall receive clear error message on failure.	In Review
FRS-015-02	Validate sensor data range	Backend API	High	Result must be visible in mobile UI within expected time.	Pending QA
FRS-015-03	Support caregiver view	Bluetooth Sensor	Not ranked	TBD by clinical SME - intentionally incomplete.	Rejected once
FRS-015-04	Generate high/low heart-rate alert	Android/iOS	Not ranked	User shall receive clear error message on failure.	Pending QA

FRS Section 16 - UI behavior

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Acceptance: expected result to be confirmed by QA; evidence screenshot may be attached later.

- Old requirement copied from web portal project - review if still valid.
- FRS-016-3: App shall process heart rate data and preserve timestamp/timezone context.
- Acceptance: expected result to be confirmed by QA; evidence screenshot may be attached later.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
FRS_16.1	Validate sensor data range	Notification Service	Not ranked	TBD by clinical SME - intentionally incomplete.	Pending QA
FRS-016-02	Export readings as PDF/CSV	Notification Service	Not ranked	Data shall be encrypted in transit and at rest.	Approved?
FRS-016-03	Support caregiver view	Notification Service	Not ranked	Data shall be encrypted in transit and at rest.	Approved?

FRS Section 17 - Risk and control mapping

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- For iOS only? For Android only? unclear; parser should capture platform if visible.
- Unnumbered note - check sensor disconnect behavior during active monitoring.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
FRS-017-01	Manual heart rate entry when device unavailable	Android/iOS	Medium	Audit/event record shall be generated where applicable.	Pending QA
FRS-017-02	Display current BPM with timestamp	Android	Low	Result must be visible in mobile UI within expected time.	Pending QA
FRS-017-03	Sync readings to cloud API	Android	Medium	User shall receive clear error message on failure.	Approved?
FRS-017-04	Handle offline mode	Cloud Sync	Not ranked	User shall receive clear error message on failure.	Draft
FRS-017-05	Battery and Bluetooth warning	iOS	Low	Result must be visible in mobile UI within expected time.	Pending QA

FRS Section 18 - Audit trail mapping

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13. FRS-018-1: App shall process heart rate data and preserve timestamp/timezone context.

- Acceptance: expected result to be confirmed by QA; evidence screenshot may be attached later.

14. Acceptance: expected result to be confirmed by QA; evidence screenshot may be attached later.

- Unnumbered note - check sensor disconnect behavior during active monitoring.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
FRS-018-01	Store readings securely on device	Notification Service	Low	Data shall be encrypted in transit and at rest.	Need SME
FRS-018-02	Notification retry and suppression	iOS	Not ranked	User shall receive clear error message on failure.	Pending QA
FRS-018-03	Sync readings to cloud API	Bluetooth Sensor	High	System shall complete the action without data loss.	Draft

MIXED TRACEABILITY LINE: URS-00X -> FRS?? -> DS component maybe -> IQ/OQ/PQ evidence TBD. This line is intentionally noisy.

FRS Section 19 - Test condition table

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- Acceptance: expected result to be confirmed by QA; evidence screenshot may be attached later.

Unnumbered note - check sensor disconnect behavior during active monitoring.

For iOS only? For Android only? unclear; parser should capture platform if visible.

15. Acceptance: expected result to be confirmed by QA; evidence screenshot may be attached later.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
FRS-019-01	Display current BPM with timestamp	Android/iOS	Low	Result must be visible in mobile UI within expected time.	Rejected once
FRS-019-02	Handle offline mode	iOS	Low	Data shall be encrypted in transit and at rest.	In Review
FRS_19.3	Display current BPM with timestamp	iOS	Medium	Result must be visible in mobile UI within expected time.	Need SME
FRS-019-04	Notification retry and suppression	Bluetooth Sensor	Medium	Data shall be encrypted in transit and at rest.	Approved?

FRS Section 20 - Evidence placeholder

OCR noisy line

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- FRS-020-1: App shall process heart rate data and preserve timestamp/timezone context.

FRS-020-2: App shall process heart rate data and preserve timestamp/timezone context.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
FRS-020-01	Generate high/low heart-rate alert	Notification Service	Medium	Result must be visible in mobile UI within expected time.	Need SME
FRS_20.2	Battery and Bluetooth warning	iOS	High	Result must be visible in mobile UI within expected time.	Need SME
FRS-020-03	Sync readings to cloud API	iOS	Not ranked	TBD by clinical SME - intentionally incomplete.	Approved?

FRS Section 21 - Security and privacy

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16. Unnumbered note - check sensor disconnect behavior during active monitoring.

Unnumbered note - check sensor disconnect behavior during active monitoring.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
FRS-021-01	Handle offline mode	iOS	High	User shall receive clear error message on failure.	Approved?
FRS-021-02	Sync readings to cloud API	Local DB	TBD	Result must be visible in mobile UI within expected time.	Rejected once
FRS-021-03	Biometric login option	Android	Low	TBD by clinical SME - intentionally incomplete.	Draft
FRS-021-04	Sync readings to cloud API	Cloud Sync	High	Data shall be encrypted in transit and at rest.	Pending QA
FRS-021-05	Capture heart rate from paired device	Bluetooth Sensor	High	Result must be visible in mobile UI within expected time.	Rejected once

FRS Section 22 - API behavior

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- Old requirement copied from web portal project - review if still valid.
17. Old requirement copied from web portal project - review if still valid.
- FRS-022-3: App shall process heart rate data and preserve timestamp/timezone context.
18. Unnumbered note - check sensor disconnect behavior during active monitoring.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
FRS-022-01	Export readings as PDF/CSV	Bluetooth Sensor	Medium	Data shall be encrypted in transit and at rest.	Rejected once
FRS-022-02	Validate sensor data range	Local DB	Low	Audit/event record shall be generated where applicable.	Pending QA
FRS_22.3	Handle offline mode	Notification Service	TBD	Result must be visible in mobile UI within expected time.	Need SME

FRS Section 23 - Requirement details

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FRS-023-1: App shall process heart rate data and preserve timestamp/timezone context.

19. Unnumbered note - check sensor disconnect behavior during active monitoring.

Old requirement copied from web portal project - review if still valid.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
FRS-023-01	Consent capture and privacy notice	Android/iOS	Medium	Data shall be encrypted in transit and at rest.	In Review
FRS-023-02	Show trend chart for 7/30/90 days	Backend API	Medium	Audit/event record shall be generated where applicable.	Pending QA
FRS-023-03	Display current BPM with timestamp	Backend API	Low	Audit/event record shall be generated where applicable.	Need SME
FRS-023-04	Sync readings to cloud API	Cloud Sync	High	Result must be visible in mobile UI within expected time.	Rejected once
FRS-023-05	Consent capture and privacy notice	Backend API	TBD	System shall complete the action without data loss.	Approved?

FRS Section 24 - Security and privacy

This page belongs to Functional Requirements Specification for the Heart Rate Monitor mobile application. The application supports Android and iOS users, paired heart-rate sensors, manual BPM entry, trend review, alerts, data export, and controlled access. This block intentionally includes uneven numbering, repeated labels, partial owner fields, and mixed wording so an AI traceability/RAG engine can be tested against non-perfect documents. Page marker: FRS-024. Owner: TBD. Important terms: BPM, sensor pairing, low alert, high alert, caregiver, audit trail, local cache, encrypted sync, consent, export, notification.

- For iOS only? For Android only? unclear; parser should capture platform if visible.

Unnumbered note - check sensor disconnect behavior during active monitoring.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
FRS-024-01	Show trend chart for 7/30/90 days	Backend API	TBD	Audit/event record shall be generated where applicable.	In Review
FRS_24.2	Sync readings to cloud API	iOS	Medium	TBD by clinical SME - intentionally incomplete.	Need SME
FRS_24.3	Display current BPM with timestamp	iOS	Medium	Result must be visible in mobile UI within expected time.	Rejected once
FRS_24.4	Validate sensor data range	Local DB	Low	System shall complete the action without data loss.	Draft

FRS Section 25 - Bluetooth edge cases

This page belongs to Functional Requirements Specification for the Heart Rate Monitor mobile application. The application supports Android and iOS users, paired heart-rate sensors, manual BPM entry, trend review, alerts, data export, and controlled access. This block intentionally includes uneven numbering, repeated labels, partial owner fields, and mixed wording so an AI traceability/RAG engine can be tested against non-perfect documents. Page marker: FRS-025. Owner: Product. Important terms: BPM, sensor pairing, low alert, high alert, caregiver, audit trail, local cache, encrypted sync, consent, export, notification.

20. FRS-025-1: App shall process heart rate data and preserve timestamp/timezone context.

Old requirement copied from web portal project - review if still valid.

- Unnumbered note - check sensor disconnect behavior during active monitoring.

Old requirement copied from web portal project - review if still valid.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
FRS-025-01	Export readings as PDF/CSV	Bluetooth Sensor	Medium	User shall receive clear error message on failure.	Pending QA
FRS-025-02	Sync readings to cloud API	Backend API	High	User shall receive clear error message on failure.	Pending QA
FRS_25.3	Battery and Bluetooth warning	Bluetooth Sensor	Not ranked	Data shall be encrypted in transit and at rest.	Draft
FRS_25.4	Support caregiver view	Bluetooth Sensor	Low	Audit/event record shall be generated where applicable.	Approved?
FRS-025-05	Biometric login option	Android	TBD	Result must be visible in mobile UI within expected time.	Rejected once

FRS Section 26 - Evidence placeholder

This page belongs to Functional Requirements Specification for the Heart Rate Monitor mobile application. The application supports Android and iOS users, paired heart-rate sensors, manual BPM entry, trend review, alerts, data export, and controlled access. This block intentionally includes uneven numbering, repeated labels, partial owner fields, and mixed wording so an AI traceability/RAG engine can be tested against non-perfect documents. Page marker: FRS-026. Owner: Product. Important terms: BPM, sensor pairing, low alert, high alert, caregiver, audit trail, local cache, encrypted sync, consent, export, notification.

- Old requirement copied from web portal project - review if still valid.

Unnumbered note - check sensor disconnect behavior during active monitoring.

- Unnumbered note - check sensor disconnect behavior during active monitoring.

21. For iOS only? For Android only? unclear; parser should capture platform if visible.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
FRS-026-01	Display current BPM with timestamp	Bluetooth Sensor	TBD	System shall complete the action without data loss.	Need SME
FRS-026-02	Sync readings to cloud API	Backend API	Medium	System shall complete the action without data loss.	In Review
FRS-026-03	Manual heart rate entry when device unavailable	Notification Service	Medium	TBD by clinical SME - intentionally incomplete.	Approved?
FRS-026-04	Display current BPM with timestamp	Cloud Sync	TBD	System shall complete the action without data loss.	Need SME

Random inserted note: duplicate title appears again below.

FRS Section 26 - Evidence placeholder

FRS Section 27 - Open questions

This page belongs to Functional Requirements Specification for the Heart Rate Monitor mobile application. The application supports Android and iOS users, paired heart-rate sensors, manual BPM entry, trend review, alerts, data export, and controlled access. This block intentionally includes uneven numbering, repeated labels, partial owner fields, and mixed wording so an AI traceability/RAG engine can be tested against non-perfect documents. Page marker: FRS-027. Owner: Validation. Important terms: BPM, sensor pairing, low alert, high alert, caregiver, audit trail, local cache, encrypted sync, consent, export, notification.

22. Unnumbered note - check sensor disconnect behavior during active monitoring.

- FRS-027-2: App shall process heart rate data and preserve timestamp/timezone context.
- FRS-027-3: App shall process heart rate data and preserve timestamp/timezone context.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
FRS-027-01	Capture heart rate from paired device	Notification Service	Low	Result must be visible in mobile UI within expected time.	Need SME
FRS-027-02	Validate sensor data range	Backend API	Medium	System shall complete the action without data loss.	Need SME
FRS-027-03	Battery and Bluetooth warning	Cloud Sync	Medium	User shall receive clear error message on failure.	Need SME

MIXED TRACEABILITY LINE: URS-00X -> FRS?? -> DS component maybe -> IQ/OQ/PQ evidence TBD. This line is intentionally noisy.

FRS Section 28 - Risk and control mapping

missing owner

This page belongs to Functional Requirements Specification for the Heart Rate Monitor mobile application. The application supports Android and iOS users, paired heart-rate sensors, manual BPM entry, trend review, alerts, data export, and controlled access. This block intentionally includes uneven numbering, repeated labels, partial owner fields, and mixed wording so an AI traceability/RAG engine can be tested against non-perfect documents. Page marker: FRS-028. Owner: Clinical SME. Important terms: BPM, sensor pairing, low alert, high alert, caregiver, audit trail, local cache, encrypted sync, consent, export, notification.

23. Unnumbered note - check sensor disconnect behavior during active monitoring.

Acceptance: expected result to be confirmed by QA; evidence screenshot may be attached later.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
FRS_28.1	Export readings as PDF/CSV	Local DB	High	TBD by clinical SME - intentionally incomplete.	Approved?
FRS-028-02	Show trend chart for 7/30/90 days	Local DB	Medium	Result must be visible in mobile UI within expected time.	Need SME
FRS-028-03	Handle offline mode	Android	TBD	TBD by clinical SME - intentionally incomplete.	Approved?
FRS-028-04	Biometric login option	Android	TBD	User shall receive clear error message on failure.	Pending QA

FRS Section 29 - Mobile platform variation

This page belongs to Functional Requirements Specification for the Heart Rate Monitor mobile application. The application supports Android and iOS users, paired heart-rate sensors, manual BPM entry, trend review, alerts, data export, and controlled access. This block intentionally includes uneven numbering, repeated labels, partial owner fields, and mixed wording so an AI traceability/RAG engine can be tested against non-perfect documents. Page marker: FRS-029. Owner: QA. Important terms: BPM, sensor pairing, low alert, high alert, caregiver, audit trail, local cache, encrypted sync, consent, export, notification.

- FRS-029-1: App shall process heart rate data and preserve timestamp/timezone context.

Unnumbered note - check sensor disconnect behavior during active monitoring.

For iOS only? For Android only? unclear; parser should capture platform if visible.

24. Unnumbered note - check sensor disconnect behavior during active monitoring.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
FRS-029-01	Manual heart rate entry when device unavailable	Notification Service	Not ranked	Result must be visible in mobile UI within expected time.	Need SME
FRS-029-02	Manual heart rate entry when device unavailable	Bluetooth Sensor	TBD	TBD by clinical SME - intentionally incomplete.	In Review
FRS_29.3	Generate high/low heart-rate alert	Android	Not ranked	Audit/event record shall be generated where applicable.	Draft
FRS-029-04	Export readings as PDF/CSV	Notification Service	Low	System shall complete the action without data loss.	In Review

FRS Section 30 - Alert workflow

This page belongs to Functional Requirements Specification for the Heart Rate Monitor mobile application. The application supports Android and iOS users, paired heart-rate sensors, manual BPM entry, trend review, alerts, data export, and controlled access. This block intentionally includes uneven numbering, repeated labels, partial owner fields, and mixed wording so an AI traceability/RAG engine can be tested against non-perfect documents. Page marker: FRS-030. Owner: Clinical SME. Important terms: BPM, sensor pairing, low alert, high alert, caregiver, audit trail, local cache, encrypted sync, consent, export, notification.

- For iOS only? For Android only? unclear; parser should capture platform if visible.

FRS-030-2: App shall process heart rate data and preserve timestamp/timezone context.

- Unnumbered note - check sensor disconnect behavior during active monitoring.
- FRS-030-4: App shall process heart rate data and preserve timestamp/timezone context.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
FRS-030-01	Battery and Bluetooth warning	Android/iOS	High	User shall receive clear error message on failure.	Draft
FRS_30.2	Audit user actions	Local DB	Medium	TBD by clinical SME - intentionally incomplete.	Need SME
FRS_30.3	Manual heart rate entry when device unavailable	Backend API	TBD	System shall complete the action without data loss.	Approved?
FRS-030-04	Support caregiver view	Cloud Sync	TBD	System shall complete the action without data loss.	Need SME
FRS-030-05	Show trend chart for 7/30/90 days	Local DB	Not ranked	Audit/event record shall be generated where applicable.	Draft

FRS Section 31 - Requirement details

This page belongs to Functional Requirements Specification for the Heart Rate Monitor mobile application. The application supports Android and iOS users, paired heart-rate sensors, manual BPM entry, trend review, alerts, data export, and controlled access. This block intentionally includes uneven numbering, repeated labels, partial owner fields, and mixed wording so an AI traceability/RAG engine can be tested against non-perfect documents. Page marker: FRS-031. Owner: QA. Important terms: BPM, sensor pairing, low alert, high alert, caregiver, audit trail, local cache, encrypted sync, consent, export, notification.

- For iOS only? For Android only? unclear; parser should capture platform if visible.
25. FRS-031-2: App shall process heart rate data and preserve timestamp/timezone context.
26. For iOS only? For Android only? unclear; parser should capture platform if visible.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
FRS-031-01	Sync readings to cloud API	Cloud Sync	TBD	TBD by clinical SME - intentionally incomplete.	Approved?
FRS-031-02	Biometric login option	iOS	High	User shall receive clear error message on failure.	Draft
FRS-031-03	Consent capture and privacy notice	Backend API	Not ranked	TBD by clinical SME - intentionally incomplete.	Approved?

FRS Section 32 - Risk and control mapping

This page belongs to Functional Requirements Specification for the Heart Rate Monitor mobile application. The application supports Android and iOS users, paired heart-rate sensors, manual BPM entry, trend review, alerts, data export, and controlled access. This block intentionally includes uneven numbering, repeated labels, partial owner fields, and mixed wording so an AI traceability/RAG engine can be tested against non-perfect documents. Page marker: FRS-032. Owner: Mobile Lead. Important terms: BPM, sensor pairing, low alert, high alert, caregiver, audit trail, local cache, encrypted sync, consent, export, notification.

- Old requirement copied from web portal project - review if still valid.

27. FRS-032-2: App shall process heart rate data and preserve timestamp/timezone context.

- FRS-032-3: App shall process heart rate data and preserve timestamp/timezone context.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
FRS-032-01	Biometric login option	Backend API	Medium	Data shall be encrypted in transit and at rest.	Draft
FRS_32.2	Notification retry and suppression	Local DB	Not ranked	Result must be visible in mobile UI within expected time.	Approved?
FRS-032-03	Generate high/low heart-rate alert	iOS	Not ranked	User shall receive clear error message on failure.	Need SME
FRS-032-04	Export readings as PDF/CSV	iOS	Not ranked	Audit/event record shall be generated where applicable.	Rejected once
FRS_32.5	Show trend chart for 7/30/90 days	Notification Service	TBD	Audit/event record shall be generated where applicable.	Pending QA

FRS Section 33 - Traceability notes

v0.1/v1.0 mixed

This page belongs to Functional Requirements Specification for the Heart Rate Monitor mobile application. The application supports Android and iOS users, paired heart-rate sensors, manual BPM entry, trend review, alerts, data export, and controlled access. This block intentionally includes uneven numbering, repeated labels, partial owner fields, and mixed wording so an AI traceability/RAG engine can be tested against non-perfect documents. Page marker: FRS-033. Owner: Validation. Important terms: BPM, sensor pairing, low alert, high alert, caregiver, audit trail, local cache, encrypted sync, consent, export, notification.

- Old requirement copied from web portal project - review if still valid.

28. Acceptance: expected result to be confirmed by QA; evidence screenshot may be attached later.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
FRS-033-01	Store readings securely on device	Bluetooth Sensor	TBD	Result must be visible in mobile UI within expected time.	Rejected once
FRS-033-02	Sync readings to cloud API	Bluetooth Sensor	TBD	System shall complete the action without data loss.	Approved?
FRS-033-03	Capture heart rate from paired device	Local DB	TBD	User shall receive clear error message on failure.	Need SME
FRS-033-04	Validate sensor data range	Android/iOS	TBD	TBD by clinical SME - intentionally incomplete.	Approved?

FRS Section 34 - Test condition table

This page belongs to Functional Requirements Specification for the Heart Rate Monitor mobile application. The application supports Android and iOS users, paired heart-rate sensors, manual BPM entry, trend review, alerts, data export, and controlled access. This block intentionally includes uneven numbering, repeated labels, partial owner fields, and mixed wording so an AI traceability/RAG engine can be tested against non-perfect documents. Page marker: FRS-034. Owner: Mobile Lead. Important terms: BPM, sensor pairing, low alert, high alert, caregiver, audit trail, local cache, encrypted sync, consent, export, notification.

FRS-034-1: App shall process heart rate data and preserve timestamp/timezone context.

FRS-034-2: App shall process heart rate data and preserve timestamp/timezone context.

Acceptance: expected result to be confirmed by QA; evidence screenshot may be attached later.

29. Old requirement copied from web portal project - review if still valid.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
FRS-034-01	Manual heart rate entry when device unavailable	Cloud Sync	Not ranked	User shall receive clear error message on failure.	Approved?
FRS-034-02	Manual heart rate entry when device unavailable	Bluetooth Sensor	Medium	System shall complete the action without data loss.	Draft
FRS-034-03	Support caregiver view	Android	TBD	Data shall be encrypted in transit and at rest.	Rejected once

FRS Section 35 - Alert workflow

This page belongs to Functional Requirements Specification for the Heart Rate Monitor mobile application. The application supports Android and iOS users, paired heart-rate sensors, manual BPM entry, trend review, alerts, data export, and controlled access. This block intentionally includes uneven numbering, repeated labels, partial owner fields, and mixed wording so an AI traceability/RAG engine can be tested against non-perfect documents. Page marker: FRS-035. Owner: Validation. Important terms: BPM, sensor pairing, low alert, high alert, caregiver, audit trail, local cache, encrypted sync, consent, export, notification.

- Old requirement copied from web portal project - review if still valid.
30. Unnumbered note - check sensor disconnect behavior during active monitoring.
31. For iOS only? For Android only? unclear; parser should capture platform if visible.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
FRS-035-01	Show trend chart for 7/30/90 days	Android	Low	TBD by clinical SME - intentionally incomplete.	Need SME
FRS-035-02	Handle offline mode	Android/iOS	Medium	System shall complete the action without data loss.	Need SME
FRS-035-03	Manual heart rate entry when device unavailable	Local DB	High	System shall complete the action without data loss.	Draft

FRS Section 36 - Traceability notes

This page belongs to Functional Requirements Specification for the Heart Rate Monitor mobile application. The application supports Android and iOS users, paired heart-rate sensors, manual BPM entry, trend review, alerts, data export, and controlled access. This block intentionally includes uneven numbering, repeated labels, partial owner fields, and mixed wording so an AI traceability/RAG engine can be tested against non-perfect documents. Page marker: FRS-036. Owner: QA. Important terms: BPM, sensor pairing, low alert, high alert, caregiver, audit trail, local cache, encrypted sync, consent, export, notification.

- FRS-036-1: App shall process heart rate data and preserve timestamp/timezone context.

For iOS only? For Android only? unclear; parser should capture platform if visible.

For iOS only? For Android only? unclear; parser should capture platform if visible.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
FRS-036-01	Handle offline mode	iOS	High	Result must be visible in mobile UI within expected time.	Approved?
FRS-036-02	Battery and Bluetooth warning	Android/iOS	TBD	System shall complete the action without data loss.	Rejected once
FRS-036-03	Export readings as PDF/CSV	iOS	TBD	TBD by clinical SME - intentionally incomplete.	In Review
FRS-036-04	Consent capture and privacy notice	Notification Service	Low	Result must be visible in mobile UI within expected time.	Pending QA

MIXED TRACEABILITY LINE: URS-00X -> FRS?? -> DS component maybe -> IQ/OQ/PQ evidence TBD. This line is intentionally noisy.

FRS Section 37 - Security and privacy

This page belongs to Functional Requirements Specification for the Heart Rate Monitor mobile application. The application supports Android and iOS users, paired heart-rate sensors, manual BPM entry, trend review, alerts, data export, and controlled access. This block intentionally includes uneven numbering, repeated labels, partial owner fields, and mixed wording so an AI traceability/RAG engine can be tested against non-perfect documents. Page marker: FRS-037. Owner: Mobile Lead. Important terms: BPM, sensor pairing, low alert, high alert, caregiver, audit trail, local cache, encrypted sync, consent, export, notification.

32. Unnumbered note - check sensor disconnect behavior during active monitoring.

- Old requirement copied from web portal project - review if still valid.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
FRS-037-01	Generate high/low heart-rate alert	Android	Not ranked	Data shall be encrypted in transit and at rest.	Rejected once
FRS-037-02	Validate sensor data range	Android/iOS	TBD	User shall receive clear error message on failure.	Pending QA
FRS-037-03	Handle offline mode	Bluetooth Sensor	Medium	System shall complete the action without data loss.	Need SME
FRS-037-04	Store readings securely on device	Android	TBD	Data shall be encrypted in transit and at rest.	Approved?

FRS Section 38 - Open questions

This page belongs to Functional Requirements Specification for the Heart Rate Monitor mobile application. The application supports Android and iOS users, paired heart-rate sensors, manual BPM entry, trend review, alerts, data export, and controlled access. This block intentionally includes uneven numbering, repeated labels, partial owner fields, and mixed wording so an AI traceability/RAG engine can be tested against non-perfect documents. Page marker: FRS-038. Owner: QA. Important terms: BPM, sensor pairing, low alert, high alert, caregiver, audit trail, local cache, encrypted sync, consent, export, notification.

- FRS-038-1: App shall process heart rate data and preserve timestamp/timezone context.
- Acceptance: expected result to be confirmed by QA; evidence screenshot may be attached later.

33. FRS-038-3: App shall process heart rate data and preserve timestamp/timezone context.

34. For iOS only? For Android only? unclear; parser should capture platform if visible.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
FRS-038-01	Notification retry and suppression	Notification Service	Not ranked	Result must be visible in mobile UI within expected time.	Rejected once
FRS-038-02	Export readings as PDF/CSV	Notification Service	Low	Audit/event record shall be generated where applicable.	Rejected once
FRS-038-03	Battery and Bluetooth warning	Android/iOS	TBD	System shall complete the action without data loss.	In Review

FRS Section 39 - Security and privacy

REVIEW-LATER

This page belongs to Functional Requirements Specification for the Heart Rate Monitor mobile application. The application supports Android and iOS users, paired heart-rate sensors, manual BPM entry, trend review, alerts, data export, and controlled access. This block intentionally includes uneven numbering, repeated labels, partial owner fields, and mixed wording so an AI traceability/RAG engine can be tested against non-perfect documents. Page marker: FRS-039. Owner: Clinical SME. Important terms: BPM, sensor pairing, low alert, high alert, caregiver, audit trail, local cache, encrypted sync, consent, export, notification.

- FRS-039-1: App shall process heart rate data and preserve timestamp/timezone context.

For iOS only? For Android only? unclear; parser should capture platform if visible.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
FRS_39.1	Validate sensor data range	Bluetooth Sensor	High	Data shall be encrypted in transit and at rest.	Pending QA
FRS-039-02	Notification retry and suppression	Notification Service	TBD	Data shall be encrypted in transit and at rest.	In Review
FRS-039-03	Generate high/low heart-rate alert	Android	TBD	Audit/event record shall be generated where applicable.	In Review
FRS-039-04	Audit user actions	Notification Service	TBD	System shall complete the action without data loss.	Rejected once

Random inserted note: duplicate title appears again below.

FRS Section 39 - Security and privacy

FRS Section 40 - Risk and control mapping

missing owner

This page belongs to Functional Requirements Specification for the Heart Rate Monitor mobile application. The application supports Android and iOS users, paired heart-rate sensors, manual BPM entry, trend review, alerts, data export, and controlled access. This block intentionally includes uneven numbering, repeated labels, partial owner fields, and mixed wording so an AI traceability/RAG engine can be tested against non-perfect documents. Page marker: FRS-040. Owner: Validation. Important terms: BPM, sensor pairing, low alert, high alert, caregiver, audit trail, local cache, encrypted sync, consent, export, notification.

- FRS-040-1: App shall process heart rate data and preserve timestamp/timezone context.

35. For iOS only? For Android only? unclear; parser should capture platform if visible.

Acceptance: expected result to be confirmed by QA; evidence screenshot may be attached later.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
FRS-040-01	Capture heart rate from paired device	Backend API	TBD	System shall complete the action without data loss.	Draft
FRS-040-02	Consent capture and privacy notice	iOS	Medium	TBD by clinical SME - intentionally incomplete.	Pending QA
FRS-040-03	Sync readings to cloud API	Backend API	High	Result must be visible in mobile UI within expected time.	Draft
FRS_40.4	Capture heart rate from paired device	Bluetooth Sensor	TBD	Data shall be encrypted in transit and at rest.	Need SME
FRS_40.5	Display current BPM with timestamp	Backend API	Medium	System shall complete the action without data loss.	Approved?

FRS Section 41 - Data handling notes

manual note

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- 36. FRS-041-1: App shall process heart rate data and preserve timestamp/timezone context.
- 37. FRS-041-2: App shall process heart rate data and preserve timestamp/timezone context.

Old requirement copied from web portal project - review if still valid.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
FRS-041-01	Export readings as PDF/CSV	Local DB	TBD	Audit/event record shall be generated where applicable.	Need SME
FRS_41.2	Battery and Bluetooth warning	Android	TBD	Data shall be encrypted in transit and at rest.	Need SME
FRS-041-03	Display current BPM with timestamp	Android/iOS	Low	Result must be visible in mobile UI within expected time.	Approved?
FRS-041-04	Handle offline mode	Notification Service	Medium	Data shall be encrypted in transit and at rest.	Rejected once

FRS Section 42 - Traceability notes

duplicate section

This page belongs to Functional Requirements Specification for the Heart Rate Monitor mobile application. The application supports Android and iOS users, paired heart-rate sensors, manual BPM entry, trend review, alerts, data export, and controlled access. This block intentionally includes uneven numbering, repeated labels, partial owner fields, and mixed wording so an AI traceability/RAG engine can be tested against non-perfect documents. Page marker: FRS-042. Owner: Validation. Important terms: BPM, sensor pairing, low alert, high alert, caregiver, audit trail, local cache, encrypted sync, consent, export, notification.

- FRS-042-1: App shall process heart rate data and preserve timestamp/timezone context.
- Old requirement copied from web portal project - review if still valid.

38. FRS-042-3: App shall process heart rate data and preserve timestamp/timezone context.
39. Acceptance: expected result to be confirmed by QA; evidence screenshot may be attached later.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
FRS-042-01	Capture heart rate from paired device	iOS	High	Audit/event record shall be generated where applicable.	Rejected once
FRS-042-02	Capture heart rate from paired device	Notification Service	Medium	Data shall be encrypted in transit and at rest.	Pending QA
FRS-042-03	Notification retry and suppression	iOS	TBD	System shall complete the action without data loss.	Need SME

FRS Section 43 - Evidence placeholder

This page belongs to Functional Requirements Specification for the Heart Rate Monitor mobile application. The application supports Android and iOS users, paired heart-rate sensors, manual BPM entry, trend review, alerts, data export, and controlled access. This block intentionally includes uneven numbering, repeated labels, partial owner fields, and mixed wording so an AI traceability/RAG engine can be tested against non-perfect documents. Page marker: FRS-043. Owner: Product. Important terms: BPM, sensor pairing, low alert, high alert, caregiver, audit trail, local cache, encrypted sync, consent, export, notification.

Old requirement copied from web portal project - review if still valid.

Unnumbered note - check sensor disconnect behavior during active monitoring.

- Acceptance: expected result to be confirmed by QA; evidence screenshot may be attached later.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
FRS-043-01	Biometric login option	Notification Service	Not ranked	Data shall be encrypted in transit and at rest.	Rejected once
FRS-043-02	Generate high/low heart-rate alert	Cloud Sync	Low	User shall receive clear error message on failure.	Approved?
FRS_43.3	Sync readings to cloud API	Android/iOS	TBD	User shall receive clear error message on failure.	In Review
FRS_43.4	Generate high/low heart-rate alert	Android/iOS	Not ranked	TBD by clinical SME - intentionally incomplete.	Rejected once

FRS Section 44 - Bluetooth edge cases

This page belongs to Functional Requirements Specification for the Heart Rate Monitor mobile application. The application supports Android and iOS users, paired heart-rate sensors, manual BPM entry, trend review, alerts, data export, and controlled access. This block intentionally includes uneven numbering, repeated labels, partial owner fields, and mixed wording so an AI traceability/RAG engine can be tested against non-perfect documents. Page marker: FRS-044. Owner: QA. Important terms: BPM, sensor pairing, low alert, high alert, caregiver, audit trail, local cache, encrypted sync, consent, export, notification.

- 40. Acceptance: expected result to be confirmed by QA; evidence screenshot may be attached later.
- 41. Old requirement copied from web portal project - review if still valid.

Old requirement copied from web portal project - review if still valid.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
FRS_44.1	Manual heart rate entry when device unavailable	Notification Service	Low	Result must be visible in mobile UI within expected time.	Approved?
FRS-044-02	Capture heart rate from paired device	Notification Service	High	TBD by clinical SME - intentionally incomplete.	In Review
FRS-044-03	Sync readings to cloud API	Cloud Sync	High	Data shall be encrypted in transit and at rest.	Approved?

FRS Section 45 - Repeated requirement block

This page belongs to Functional Requirements Specification for the Heart Rate Monitor mobile application. The application supports Android and iOS users, paired heart-rate sensors, manual BPM entry, trend review, alerts, data export, and controlled access. This block intentionally includes uneven numbering, repeated labels, partial owner fields, and mixed wording so an AI traceability/RAG engine can be tested against non-perfect documents. Page marker: FRS-045. Owner: QA. Important terms: BPM, sensor pairing, low alert, high alert, caregiver, audit trail, local cache, encrypted sync, consent, export, notification.

Unnumbered note - check sensor disconnect behavior during active monitoring.

- For iOS only? For Android only? unclear; parser should capture platform if visible.
- Old requirement copied from web portal project - review if still valid.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
FRS-045-01	Support caregiver view	Android	High	Result must be visible in mobile UI within expected time.	Draft
FRS-045-02	Display current BPM with timestamp	Local DB	Medium	Result must be visible in mobile UI within expected time.	Draft
FRS-045-03	Notification retry and suppression	Android	Not ranked	Audit/event record shall be generated where applicable.	Pending QA
FRS_45.4	Manual heart rate entry when device unavailable	Notification Service	High	Result must be visible in mobile UI within expected time.	Approved?
FRS-045-05	Notification retry and suppression	Bluetooth Sensor	Low	Audit/event record shall be generated where applicable.	Pending QA

MIXED TRACEABILITY LINE: URS-00X -> FRS?? -> DS component maybe -> IQ/OQ/PQ evidence TBD. This line is intentionally noisy.

FRS Section 46 - Test condition table

This page belongs to Functional Requirements Specification for the Heart Rate Monitor mobile application. The application supports Android and iOS users, paired heart-rate sensors, manual BPM entry, trend review, alerts, data export, and controlled access. This block intentionally includes uneven numbering, repeated labels, partial owner fields, and mixed wording so an AI traceability/RAG engine can be tested against non-perfect documents. Page marker: FRS-046. Owner: Validation. Important terms: BPM, sensor pairing, low alert, high alert, caregiver, audit trail, local cache, encrypted sync, consent, export, notification.

- 42. Unnumbered note - check sensor disconnect behavior during active monitoring.
- 43. FRS-046-2: App shall process heart rate data and preserve timestamp/timezone context.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
FRS-046-01	Consent capture and privacy notice	Android/iOS	Not ranked	System shall complete the action without data loss.	Rejected once
FRS-046-02	Validate sensor data range	Backend API	Not ranked	Audit/event record shall be generated where applicable.	In Review
FRS-046-03	Generate high/low heart-rate alert	Android	Medium	TBD by clinical SME - intentionally incomplete.	Rejected once
FRS-046-04	Display current BPM with timestamp	Notification Service	Not ranked	User shall receive clear error message on failure.	Draft
FRS-046-05	Export readings as PDF/CSV	Android	High	Result must be visible in mobile UI within expected time.	Draft

FRS Section 47 - Requirement details

This page belongs to Functional Requirements Specification for the Heart Rate Monitor mobile application. The application supports Android and iOS users, paired heart-rate sensors, manual BPM entry, trend review, alerts, data export, and controlled access. This block intentionally includes uneven numbering, repeated labels, partial owner fields, and mixed wording so an AI traceability/RAG engine can be tested against non-perfect documents. Page marker: FRS-047. Owner: QA. Important terms: BPM, sensor pairing, low alert, high alert, caregiver, audit trail, local cache, encrypted sync, consent, export, notification.

- FRS-047-1: App shall process heart rate data and preserve timestamp/timezone context.

Acceptance: expected result to be confirmed by QA; evidence screenshot may be attached later.

Unnumbered note - check sensor disconnect behavior during active monitoring.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
FRS_47.1	Support caregiver view	Android	TBD	TBD by clinical SME - intentionally incomplete.	Rejected once
FRS-047-02	Support caregiver view	Cloud Sync	Not ranked	System shall complete the action without data loss.	Rejected once
FRS_47.3	Show trend chart for 7/30/90 days	Android/iOS	Medium	System shall complete the action without data loss.	In Review
FRS-047-04	Battery and Bluetooth warning	Bluetooth Sensor	Low	System shall complete the action without data loss.	Draft

FRS Section 48 - Data handling notes

This page belongs to Functional Requirements Specification for the Heart Rate Monitor mobile application. The application supports Android and iOS users, paired heart-rate sensors, manual BPM entry, trend review, alerts, data export, and controlled access. This block intentionally includes uneven numbering, repeated labels, partial owner fields, and mixed wording so an AI traceability/RAG engine can be tested against non-perfect documents. Page marker: FRS-048. Owner: Product. Important terms: BPM, sensor pairing, low alert, high alert, caregiver, audit trail, local cache, encrypted sync, consent, export, notification.

- Acceptance: expected result to be confirmed by QA; evidence screenshot may be attached later.

44. FRS-048-2: App shall process heart rate data and preserve timestamp/timezone context.

45. Acceptance: expected result to be confirmed by QA; evidence screenshot may be attached later.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
FRS-048-01	Audit user actions	Android	Medium	Result must be visible in mobile UI within expected time.	Pending QA
FRS-048-02	Battery and Bluetooth warning	Android	Not ranked	Data shall be encrypted in transit and at rest.	Approved?
FRS-048-03	Store readings securely on device	Android/iOS	TBD	User shall receive clear error message on failure.	Pending QA
FRS-048-04	Notification retry and suppression	Local DB	Low	Result must be visible in mobile UI within expected time.	Approved?
FRS-048-05	Manual heart rate entry when device unavailable	Cloud Sync	Low	Audit/event record shall be generated where applicable.	Draft

FRS Section 49 - Audit trail mapping

This page belongs to Functional Requirements Specification for the Heart Rate Monitor mobile application. The application supports Android and iOS users, paired heart-rate sensors, manual BPM entry, trend review, alerts, data export, and controlled access. This block intentionally includes uneven numbering, repeated labels, partial owner fields, and mixed wording so an AI traceability/RAG engine can be tested against non-perfect documents. Page marker: FRS-049. Owner: TBD. Important terms: BPM, sensor pairing, low alert, high alert, caregiver, audit trail, local cache, encrypted sync, consent, export, notification.

- Acceptance: expected result to be confirmed by QA; evidence screenshot may be attached later.
- Old requirement copied from web portal project - review if still valid.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
FRS-049-01	Display current BPM with timestamp	Android/iOS	Medium	System shall complete the action without data loss.	Approved?
FRS-049-02	Store readings securely on device	Android	TBD	Data shall be encrypted in transit and at rest.	Approved?
FRS_49.3	Sync readings to cloud API	Cloud Sync	Medium	User shall receive clear error message on failure.	Rejected once

FRS Section 50 - Alert workflow

This page belongs to Functional Requirements Specification for the Heart Rate Monitor mobile application. The application supports Android and iOS users, paired heart-rate sensors, manual BPM entry, trend review, alerts, data export, and controlled access. This block intentionally includes uneven numbering, repeated labels, partial owner fields, and mixed wording so an AI traceability/RAG engine can be tested against non-perfect documents. Page marker: FRS-050. Owner: TBD. Important terms: BPM, sensor pairing, low alert, high alert, caregiver, audit trail, local cache, encrypted sync, consent, export, notification.

Unnumbered note - check sensor disconnect behavior during active monitoring.

FRS-050-2: App shall process heart rate data and preserve timestamp/timezone context.

46. For iOS only? For Android only? unclear; parser should capture platform if visible.

- For iOS only? For Android only? unclear; parser should capture platform if visible.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
FRS_50.1	Export readings as PDF/CSV	Android	High	Audit/event record shall be generated where applicable.	In Review
FRS-050-02	Consent capture and privacy notice	Notification Service	Medium	Result must be visible in mobile UI within expected time.	Draft
FRS-050-03	Audit user actions	Cloud Sync	Medium	Data shall be encrypted in transit and at rest.	Pending QA
FRS_50.4	Manual heart rate entry when device unavailable	Backend API	Medium	Data shall be encrypted in transit and at rest.	In Review
FRS-050-05	Display current BPM with timestamp	Android	TBD	User shall receive clear error message on failure.	Need SME

FRS Section 51 - UI behavior

v0.1/v1.0 mixed

This page belongs to Functional Requirements Specification for the Heart Rate Monitor mobile application. The application supports Android and iOS users, paired heart-rate sensors, manual BPM entry, trend review, alerts, data export, and controlled access. This block intentionally includes uneven numbering, repeated labels, partial owner fields, and mixed wording so an AI traceability/RAG engine can be tested against non-perfect documents. Page marker: FRS-051. Owner: QA. Important terms: BPM, sensor pairing, low alert, high alert, caregiver, audit trail, local cache, encrypted sync, consent, export, notification.

47. Old requirement copied from web portal project - review if still valid.

- Acceptance: expected result to be confirmed by QA; evidence screenshot may be attached later.

FRS-051-3: App shall process heart rate data and preserve timestamp/timezone context.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
FRS-051-01	Sync readings to cloud API	Local DB	Not ranked	Audit/event record shall be generated where applicable.	Need SME
FRS-051-02	Display current BPM with timestamp	Cloud Sync	TBD	TBD by clinical SME - intentionally incomplete.	Rejected once
FRS-051-03	Validate sensor data range	Local DB	TBD	System shall complete the action without data loss.	Rejected once

FRS Section 52 - Exception handling

This page belongs to Functional Requirements Specification for the Heart Rate Monitor mobile application. The application supports Android and iOS users, paired heart-rate sensors, manual BPM entry, trend review, alerts, data export, and controlled access. This block intentionally includes uneven numbering, repeated labels, partial owner fields, and mixed wording so an AI traceability/RAG engine can be tested against non-perfect documents. Page marker: FRS-052. Owner: Product. Important terms: BPM, sensor pairing, low alert, high alert, caregiver, audit trail, local cache, encrypted sync, consent, export, notification.

For iOS only? For Android only? unclear; parser should capture platform if visible.

48. Acceptance: expected result to be confirmed by QA; evidence screenshot may be attached later.

Acceptance: expected result to be confirmed by QA; evidence screenshot may be attached later.

49. Acceptance: expected result to be confirmed by QA; evidence screenshot may be attached later.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
FRS-052-01	Biometric login option	Android	Not ranked	Audit/event record shall be generated where applicable.	Pending QA
FRS-052-02	Display current BPM with timestamp	Bluetooth Sensor	Medium	Result must be visible in mobile UI within expected time.	Rejected once
FRS-052-03	Store readings securely on device	Backend API	TBD	User shall receive clear error message on failure.	Draft
FRS-052-04	Display current BPM with timestamp	Android/iOS	TBD	TBD by clinical SME - intentionally incomplete.	Draft

Random inserted note: duplicate title appears again below.

FRS Section 52 - Exception handling

FRS Section 53 - Alert workflow

old text retained

This page belongs to Functional Requirements Specification for the Heart Rate Monitor mobile application. The application supports Android and iOS users, paired heart-rate sensors, manual BPM entry, trend review, alerts, data export, and controlled access. This block intentionally includes uneven numbering, repeated labels, partial owner fields, and mixed wording so an AI traceability/RAG engine can be tested against non-perfect documents. Page marker: FRS-053. Owner: Clinical SME. Important terms: BPM, sensor pairing, low alert, high alert, caregiver, audit trail, local cache, encrypted sync, consent, export, notification.

- Acceptance: expected result to be confirmed by QA; evidence screenshot may be attached later.

For iOS only? For Android only? unclear; parser should capture platform if visible.

- Unnumbered note - check sensor disconnect behavior during active monitoring.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
FRS_53.1	Consent capture and privacy notice	Backend API	Medium	TBD by clinical SME - intentionally incomplete.	In Review
FRS_53.2	Validate sensor data range	Local DB	Low	User shall receive clear error message on failure.	Rejected once
FRS-053-03	Store readings securely on device	Notification Service	Medium	User shall receive clear error message on failure.	Pending QA
FRS-053-04	Validate sensor data range	Local DB	Medium	User shall receive clear error message on failure.	Pending QA

FRS Section 54 - Requirement details

This page belongs to Functional Requirements Specification for the Heart Rate Monitor mobile application. The application supports Android and iOS users, paired heart-rate sensors, manual BPM entry, trend review, alerts, data export, and controlled access. This block intentionally includes uneven numbering, repeated labels, partial owner fields, and mixed wording so an AI traceability/RAG engine can be tested against non-perfect documents. Page marker: FRS-054. Owner: Mobile Lead. Important terms: BPM, sensor pairing, low alert, high alert, caregiver, audit trail, local cache, encrypted sync, consent, export, notification.

50. For iOS only? For Android only? unclear; parser should capture platform if visible.

Unnumbered note - check sensor disconnect behavior during active monitoring.

Unnumbered note - check sensor disconnect behavior during active monitoring.

- Old requirement copied from web portal project - review if still valid.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
FRS-054-01	Consent capture and privacy notice	iOS	Not ranked	Result must be visible in mobile UI within expected time.	Approved?
FRS-054-02	Generate high/low heart-rate alert	iOS	TBD	TBD by clinical SME - intentionally incomplete.	In Review
FRS_54.3	Export readings as PDF/CSV	Android/iOS	TBD	Audit/event record shall be generated where applicable.	Need SME
FRS-054-04	Export readings as PDF/CSV	Notification Service	Medium	Result must be visible in mobile UI within expected time.	Pending QA

MIXED TRACEABILITY LINE: URS-00X -> FRS?? -> DS component maybe -> IQ/OQ/PQ evidence TBD. This line is intentionally noisy.

FRS Section 55 - Traceability notes

v0.1/v1.0 mixed

This page belongs to Functional Requirements Specification for the Heart Rate Monitor mobile application. The application supports Android and iOS users, paired heart-rate sensors, manual BPM entry, trend review, alerts, data export, and controlled access. This block intentionally includes uneven numbering, repeated labels, partial owner fields, and mixed wording so an AI traceability/RAG engine can be tested against non-perfect documents. Page marker: FRS-055. Owner: Validation. Important terms: BPM, sensor pairing, low alert, high alert, caregiver, audit trail, local cache, encrypted sync, consent, export, notification.

- FRS-055-1: App shall process heart rate data and preserve timestamp/timezone context.
- FRS-055-2: App shall process heart rate data and preserve timestamp/timezone context.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
FRS-055-01	Validate sensor data range	Bluetooth Sensor	TBD	System shall complete the action without data loss.	In Review
FRS-055-02	Generate high/low heart-rate alert	Cloud Sync	Medium	TBD by clinical SME - intentionally incomplete.	Draft
FRS-055-03	Show trend chart for 7/30/90 days	Cloud Sync	Medium	Result must be visible in mobile UI within expected time.	Rejected once
FRS-055-04	Show trend chart for 7/30/90 days	Android/iOS	High	System shall complete the action without data loss.	Draft

FRS Section 56 - Exception handling

This page belongs to Functional Requirements Specification for the Heart Rate Monitor mobile application. The application supports Android and iOS users, paired heart-rate sensors, manual BPM entry, trend review, alerts, data export, and controlled access. This block intentionally includes uneven numbering, repeated labels, partial owner fields, and mixed wording so an AI traceability/RAG engine can be tested against non-perfect documents. Page marker: FRS-056. Owner: QA. Important terms: BPM, sensor pairing, low alert, high alert, caregiver, audit trail, local cache, encrypted sync, consent, export, notification.

- Unnumbered note - check sensor disconnect behavior during active monitoring.
- For iOS only? For Android only? unclear; parser should capture platform if visible.
- Acceptance: expected result to be confirmed by QA; evidence screenshot may be attached later.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
FRS-056-01	Capture heart rate from paired device	Notification Service	TBD	User shall receive clear error message on failure.	Need SME
FRS-056-02	Battery and Bluetooth warning	Android	Medium	Result must be visible in mobile UI within expected time.	Draft
FRS-056-03	Biometric login option	iOS	High	System shall complete the action without data loss.	Need SME
FRS-056-04	Sync readings to cloud API	Cloud Sync	High	TBD by clinical SME - intentionally incomplete.	Need SME

FRS Section 57 - Traceability notes

This page belongs to Functional Requirements Specification for the Heart Rate Monitor mobile application. The application supports Android and iOS users, paired heart-rate sensors, manual BPM entry, trend review, alerts, data export, and controlled access. This block intentionally includes uneven numbering, repeated labels, partial owner fields, and mixed wording so an AI traceability/RAG engine can be tested against non-perfect documents. Page marker: FRS-057. Owner: QA. Important terms: BPM, sensor pairing, low alert, high alert, caregiver, audit trail, local cache, encrypted sync, consent, export, notification.

51. Acceptance: expected result to be confirmed by QA; evidence screenshot may be attached later.

- Unnumbered note - check sensor disconnect behavior during active monitoring.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
FRS_57.1	Export readings as PDF/CSV	Local DB	Low	System shall complete the action without data loss.	Pending QA
FRS-057-02	Audit user actions	Bluetooth Sensor	Not ranked	User shall receive clear error message on failure.	Need SME
FRS_57.3	Generate high/low heart-rate alert	iOS	Not ranked	System shall complete the action without data loss.	Need SME
FRS-057-04	Support caregiver view	Android	TBD	System shall complete the action without data loss.	Need SME
FRS-057-05	Store readings securely on device	Android	Not ranked	User shall receive clear error message on failure.	Pending QA

FRS Section 58 - Risk and control mapping

v0.1/v1.0 mixed

This page belongs to Functional Requirements Specification for the Heart Rate Monitor mobile application. The application supports Android and iOS users, paired heart-rate sensors, manual BPM entry, trend review, alerts, data export, and controlled access. This block intentionally includes uneven numbering, repeated labels, partial owner fields, and mixed wording so an AI traceability/RAG engine can be tested against non-perfect documents. Page marker: FRS-058. Owner: TBD. Important terms: BPM, sensor pairing, low alert, high alert, caregiver, audit trail, local cache, encrypted sync, consent, export, notification.

Old requirement copied from web portal project - review if still valid.

52. Unnumbered note - check sensor disconnect behavior during active monitoring.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
FRS_58.1	Consent capture and privacy notice	Android	TBD	Data shall be encrypted in transit and at rest.	Rejected once
FRS-058-02	Consent capture and privacy notice	Cloud Sync	Not ranked	Result must be visible in mobile UI within expected time.	Approved?
FRS-058-03	Notification retry and suppression	Cloud Sync	Medium	Data shall be encrypted in transit and at rest.	Approved?
FRS-058-04	Manual heart rate entry when device unavailable	Local DB	Not ranked	System shall complete the action without data loss.	Approved?

FRS Section 59 - Data handling notes

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53. Unnumbered note - check sensor disconnect behavior during active monitoring.

For iOS only? For Android only? unclear; parser should capture platform if visible.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
FRS-059-01	Export readings as PDF/CSV	Backend API	Medium	System shall complete the action without data loss.	Rejected once
FRS-059-02	Sync readings to cloud API	Android/iOS	TBD	User shall receive clear error message on failure.	Pending QA
FRS-059-03	Support caregiver view	Cloud Sync	Not ranked	Audit/event record shall be generated where applicable.	Rejected once

FRS Section 60 - Traceability notes

manual note

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Unnumbered note - check sensor disconnect behavior during active monitoring.

Unnumbered note - check sensor disconnect behavior during active monitoring.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
FRS-060-01	Show trend chart for 7/30/90 days	Backend API	Medium	Audit/event record shall be generated where applicable.	Approved?
FRS-060-02	Manual heart rate entry when device unavailable	Backend API	Not ranked	System shall complete the action without data loss.	Approved?
FRS-060-03	Store readings securely on device	Local DB	TBD	Result must be visible in mobile UI within expected time.	Approved?
FRS-060-04	Support caregiver view	Android/iOS	Not ranked	Data shall be encrypted in transit and at rest.	Draft

FRS Section 61 - Exception handling

This page belongs to Functional Requirements Specification for the Heart Rate Monitor mobile application. The application supports Android and iOS users, paired heart-rate sensors, manual BPM entry, trend review, alerts, data export, and controlled access. This block intentionally includes uneven numbering, repeated labels, partial owner fields, and mixed wording so an AI traceability/RAG engine can be tested against non-perfect documents. Page marker: FRS-061. Owner: TBD. Important terms: BPM, sensor pairing, low alert, high alert, caregiver, audit trail, local cache, encrypted sync, consent, export, notification.

54. Unnumbered note - check sensor disconnect behavior during active monitoring.

- Acceptance: expected result to be confirmed by QA; evidence screenshot may be attached later.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
FRS-061-01	Consent capture and privacy notice	Backend API	High	TBD by clinical SME - intentionally incomplete.	Approved?
FRS-061-02	Notification retry and suppression	Backend API	Not ranked	User shall receive clear error message on failure.	Approved?
FRS-061-03	Consent capture and privacy notice	Local DB	TBD	Result must be visible in mobile UI within expected time.	In Review
FRS_61.4	Generate high/low heart-rate alert	Android	Low	Result must be visible in mobile UI within expected time.	Need SME

FRS Section 62 - Evidence placeholder

This page belongs to Functional Requirements Specification for the Heart Rate Monitor mobile application. The application supports Android and iOS users, paired heart-rate sensors, manual BPM entry, trend review, alerts, data export, and controlled access. This block intentionally includes uneven numbering, repeated labels, partial owner fields, and mixed wording so an AI traceability/RAG engine can be tested against non-perfect documents. Page marker: FRS-062. Owner: Validation. Important terms: BPM, sensor pairing, low alert, high alert, caregiver, audit trail, local cache, encrypted sync, consent, export, notification.

Acceptance: expected result to be confirmed by QA; evidence screenshot may be attached later.

55. Unnumbered note - check sensor disconnect behavior during active monitoring.

- For iOS only? For Android only? unclear; parser should capture platform if visible.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
FRS-062-01	Notification retry and suppression	Backend API	High	TBD by clinical SME - intentionally incomplete.	In Review
FRS-062-02	Generate high/low heart-rate alert	Backend API	Low	TBD by clinical SME - intentionally incomplete.	Need SME
FRS_62.3	Notification retry and suppression	Bluetooth Sensor	Low	Result must be visible in mobile UI within expected time.	Approved?

FRS Section 63 - Evidence placeholder

This page belongs to Functional Requirements Specification for the Heart Rate Monitor mobile application. The application supports Android and iOS users, paired heart-rate sensors, manual BPM entry, trend review, alerts, data export, and controlled access. This block intentionally includes uneven numbering, repeated labels, partial owner fields, and mixed wording so an AI traceability/RAG engine can be tested against non-perfect documents. Page marker: FRS-063. Owner: Clinical SME. Important terms: BPM, sensor pairing, low alert, high alert, caregiver, audit trail, local cache, encrypted sync, consent, export, notification.

56. Unnumbered note - check sensor disconnect behavior during active monitoring.

Unnumbered note - check sensor disconnect behavior during active monitoring.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
FRS_63.1	Biometric login option	Backend API	Low	Audit/event record shall be generated where applicable.	Draft
FRS-063-02	Notification retry and suppression	Android	Medium	Audit/event record shall be generated where applicable.	Approved?
FRS-063-03	Handle offline mode	Android	Low	TBD by clinical SME - intentionally incomplete.	Draft

MIXED TRACEABILITY LINE: URS-00X -> FRS?? -> DS component maybe -> IQ/OQ/PQ evidence TBD. This line is intentionally noisy.

FRS Section 64 - Exception handling

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57. Unnumbered note - check sensor disconnect behavior during active monitoring.

- Old requirement copied from web portal project - review if still valid.
- Unnumbered note - check sensor disconnect behavior during active monitoring.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
FRS_64.1	Handle offline mode	Cloud Sync	Medium	System shall complete the action without data loss.	Pending QA
FRS-064-02	Manual heart rate entry when device unavailable	Bluetooth Sensor	TBD	Audit/event record shall be generated where applicable.	Rejected once
FRS-064-03	Display current BPM with timestamp	Cloud Sync	Low	TBD by clinical SME - intentionally incomplete.	Pending QA
FRS-064-04	Capture heart rate from paired device	iOS	Medium	TBD by clinical SME - intentionally incomplete.	Rejected once

FRS Section 65 - Open questions

v0.1/v1.0 mixed

This page belongs to Functional Requirements Specification for the Heart Rate Monitor mobile application. The application supports Android and iOS users, paired heart-rate sensors, manual BPM entry, trend review, alerts, data export, and controlled access. This block intentionally includes uneven numbering, repeated labels, partial owner fields, and mixed wording so an AI traceability/RAG engine can be tested against non-perfect documents. Page marker: FRS-065. Owner: Validation. Important terms: BPM, sensor pairing, low alert, high alert, caregiver, audit trail, local cache, encrypted sync, consent, export, notification.

- For iOS only? For Android only? unclear; parser should capture platform if visible.
- Acceptance: expected result to be confirmed by QA; evidence screenshot may be attached later.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
FRS-065-01	Handle offline mode	Android	Low	Audit/event record shall be generated where applicable.	Draft
FRS-065-02	Validate sensor data range	Android	Low	System shall complete the action without data loss.	Rejected once
FRS_65.3	Battery and Bluetooth warning	Android/iOS	TBD	Audit/event record shall be generated where applicable.	Need SME
FRS-065-04	Store readings securely on device	Android/iOS	Medium	TBD by clinical SME - intentionally incomplete.	Rejected once

Random inserted note: duplicate title appears again below.

FRS Section 65 - Open questions

FRS Section 66 - Mobile platform variation

??

This page belongs to Functional Requirements Specification for the Heart Rate Monitor mobile application. The application supports Android and iOS users, paired heart-rate sensors, manual BPM entry, trend review, alerts, data export, and controlled access. This block intentionally includes uneven numbering, repeated labels, partial owner fields, and mixed wording so an AI traceability/RAG engine can be tested against non-perfect documents. Page marker: FRS-066. Owner: Mobile Lead. Important terms: BPM, sensor pairing, low alert, high alert, caregiver, audit trail, local cache, encrypted sync, consent, export, notification.

- FRS-066-1: App shall process heart rate data and preserve timestamp/timezone context.
- FRS-066-2: App shall process heart rate data and preserve timestamp/timezone context.

Unnumbered note - check sensor disconnect behavior during active monitoring.

Unnumbered note - check sensor disconnect behavior during active monitoring.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
FRS-066-01	Manual heart rate entry when device unavailable	Android/iOS	Not ranked	Result must be visible in mobile UI within expected time.	Rejected once
FRS-066-02	Biometric login option	Local DB	Not ranked	System shall complete the action without data loss.	Need SME
FRS-066-03	Store readings securely on device	iOS	High	Data shall be encrypted in transit and at rest.	Pending QA
FRS-066-04	Generate high/low heart-rate alert	Android/iOS	Low	Data shall be encrypted in transit and at rest.	Approved?
FRS-066-05	Battery and Bluetooth warning	Cloud Sync	High	User shall receive clear error message on failure.	Need SME

FRS Section 67 - Repeated requirement block

This page belongs to Functional Requirements Specification for the Heart Rate Monitor mobile application. The application supports Android and iOS users, paired heart-rate sensors, manual BPM entry, trend review, alerts, data export, and controlled access. This block intentionally includes uneven numbering, repeated labels, partial owner fields, and mixed wording so an AI traceability/RAG engine can be tested against non-perfect documents. Page marker: FRS-067. Owner: Mobile Lead. Important terms: BPM, sensor pairing, low alert, high alert, caregiver, audit trail, local cache, encrypted sync, consent, export, notification.

For iOS only? For Android only? unclear; parser should capture platform if visible.

- For iOS only? For Android only? unclear; parser should capture platform if visible.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
FRS-067-01	Consent capture and privacy notice	Bluetooth Sensor	Low	Data shall be encrypted in transit and at rest.	Pending QA
FRS-067-02	Show trend chart for 7/30/90 days	Cloud Sync	TBD	Audit/event record shall be generated where applicable.	Approved?
FRS-067-03	Generate high/low heart-rate alert	Cloud Sync	Not ranked	User shall receive clear error message on failure.	In Review

FRS Section 68 - Exception handling

This page belongs to Functional Requirements Specification for the Heart Rate Monitor mobile application. The application supports Android and iOS users, paired heart-rate sensors, manual BPM entry, trend review, alerts, data export, and controlled access. This block intentionally includes uneven numbering, repeated labels, partial owner fields, and mixed wording so an AI traceability/RAG engine can be tested against non-perfect documents. Page marker: FRS-068. Owner: Mobile Lead. Important terms: BPM, sensor pairing, low alert, high alert, caregiver, audit trail, local cache, encrypted sync, consent, export, notification.

58. Old requirement copied from web portal project - review if still valid.

Old requirement copied from web portal project - review if still valid.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
FRS-068-01	Store readings securely on device	Backend API	Medium	Result must be visible in mobile UI within expected time.	Approved?
FRS_68.2	Handle offline mode	Android/iOS	Not ranked	System shall complete the action without data loss.	Draft
FRS_68.3	Export readings as PDF/CSV	Notification Service	Medium	TBD by clinical SME - intentionally incomplete.	Need SME
FRS-068-04	Show trend chart for 7/30/90 days	Cloud Sync	TBD	Result must be visible in mobile UI within expected time.	Need SME
FRS-068-05	Generate high/low heart-rate alert	Local DB	Not ranked	Audit/event record shall be generated where applicable.	Approved?

FRS Section 69 - Data handling notes

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- Old requirement copied from web portal project - review if still valid.
- For iOS only? For Android only? unclear; parser should capture platform if visible.

59. FRS-069-3: App shall process heart rate data and preserve timestamp/timezone context.

For iOS only? For Android only? unclear; parser should capture platform if visible.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
FRS-069-01	Support caregiver view	Android	High	System shall complete the action without data loss.	Draft
FRS-069-02	Generate high/low heart-rate alert	Android	TBD	Data shall be encrypted in transit and at rest.	Approved?
FRS-069-03	Manual heart rate entry when device unavailable	Cloud Sync	Low	User shall receive clear error message on failure.	Approved?

FRS Section 70 - Alert workflow

This page belongs to Functional Requirements Specification for the Heart Rate Monitor mobile application. The application supports Android and iOS users, paired heart-rate sensors, manual BPM entry, trend review, alerts, data export, and controlled access. This block intentionally includes uneven numbering, repeated labels, partial owner fields, and mixed wording so an AI traceability/RAG engine can be tested against non-perfect documents. Page marker: FRS-070. Owner: Mobile Lead. Important terms: BPM, sensor pairing, low alert, high alert, caregiver, audit trail, local cache, encrypted sync, consent, export, notification.

For iOS only? For Android only? unclear; parser should capture platform if visible.

Acceptance: expected result to be confirmed by QA; evidence screenshot may be attached later.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
FRS-070-01	Notification retry and suppression	iOS	Low	TBD by clinical SME - intentionally incomplete.	Need SME
FRS-070-02	Export readings as PDF/CSV	Notification Service	High	User shall receive clear error message on failure.	Rejected once
FRS-070-03	Battery and Bluetooth warning	Android/iOS	TBD	TBD by clinical SME - intentionally incomplete.	Approved?
FRS-070-04	Biometric login option	Android	Medium	Result must be visible in mobile UI within expected time.	Rejected once

FRS Section 71 - Bluetooth edge cases

OCR noisy line

This page belongs to Functional Requirements Specification for the Heart Rate Monitor mobile application. The application supports Android and iOS users, paired heart-rate sensors, manual BPM entry, trend review, alerts, data export, and controlled access. This block intentionally includes uneven numbering, repeated labels, partial owner fields, and mixed wording so an AI traceability/RAG engine can be tested against non-perfect documents. Page marker: FRS-071. Owner: QA. Important terms: BPM, sensor pairing, low alert, high alert, caregiver, audit trail, local cache, encrypted sync, consent, export, notification.

Old requirement copied from web portal project - review if still valid.

- Old requirement copied from web portal project - review if still valid.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
FRS-071-01	Sync readings to cloud API	Backend API	TBD	System shall complete the action without data loss.	Draft
FRS_71.2	Audit user actions	Cloud Sync	Not ranked	Audit/event record shall be generated where applicable.	Approved?
FRS-071-03	Audit user actions	Bluetooth Sensor	Medium	Result must be visible in mobile UI within expected time.	Rejected once

FRS Section 72 - Data handling notes

This page belongs to Functional Requirements Specification for the Heart Rate Monitor mobile application. The application supports Android and iOS users, paired heart-rate sensors, manual BPM entry, trend review, alerts, data export, and controlled access. This block intentionally includes uneven numbering, repeated labels, partial owner fields, and mixed wording so an AI traceability/RAG engine can be tested against non-perfect documents. Page marker: FRS-072. Owner: Product. Important terms: BPM, sensor pairing, low alert, high alert, caregiver, audit trail, local cache, encrypted sync, consent, export, notification.

- Unnumbered note - check sensor disconnect behavior during active monitoring.

60. Acceptance: expected result to be confirmed by QA; evidence screenshot may be attached later.

61. Old requirement copied from web portal project - review if still valid.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
FRS-072-01	Audit user actions	Bluetooth Sensor	Low	Data shall be encrypted in transit and at rest.	Pending QA
FRS-072-02	Battery and Bluetooth warning	Cloud Sync	Not ranked	TBD by clinical SME - intentionally incomplete.	Draft
FRS_72.3	Display current BPM with timestamp	Local DB	Not ranked	TBD by clinical SME - intentionally incomplete.	In Review

MIXED TRACEABILITY LINE: URS-00X -> FRS?? -> DS component maybe -> IQ/OQ/PQ evidence TBD. This line is intentionally noisy.

FRS Section 73 - Audit trail mapping

v0.1/v1.0 mixed

This page belongs to Functional Requirements Specification for the Heart Rate Monitor mobile application. The application supports Android and iOS users, paired heart-rate sensors, manual BPM entry, trend review, alerts, data export, and controlled access. This block intentionally includes uneven numbering, repeated labels, partial owner fields, and mixed wording so an AI traceability/RAG engine can be tested against non-perfect documents. Page marker: FRS-073. Owner: Mobile Lead. Important terms: BPM, sensor pairing, low alert, high alert, caregiver, audit trail, local cache, encrypted sync, consent, export, notification.

Unnumbered note - check sensor disconnect behavior during active monitoring.

Unnumbered note - check sensor disconnect behavior during active monitoring.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
FRS-073-01	Show trend chart for 7/30/90 days	Local DB	Low	User shall receive clear error message on failure.	Pending QA
FRS-073-02	Notification retry and suppression	Cloud Sync	Not ranked	System shall complete the action without data loss.	Rejected once
FRS-073-03	Biometric login option	Local DB	High	Result must be visible in mobile UI within expected time.	Draft
FRS_73.4	Sync readings to cloud API	Android	TBD	Result must be visible in mobile UI within expected time.	Need SME

FRS Section 74 - Security and privacy

This page belongs to Functional Requirements Specification for the Heart Rate Monitor mobile application. The application supports Android and iOS users, paired heart-rate sensors, manual BPM entry, trend review, alerts, data export, and controlled access. This block intentionally includes uneven numbering, repeated labels, partial owner fields, and mixed wording so an AI traceability/RAG engine can be tested against non-perfect documents. Page marker: FRS-074. Owner: Mobile Lead. Important terms: BPM, sensor pairing, low alert, high alert, caregiver, audit trail, local cache, encrypted sync, consent, export, notification.

62. Old requirement copied from web portal project - review if still valid.

For iOS only? For Android only? unclear; parser should capture platform if visible.

Old requirement copied from web portal project - review if still valid.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
FRS-074-01	Generate high/low heart-rate alert	Android	TBD	TBD by clinical SME - intentionally incomplete.	In Review
FRS_74.2	Generate high/low heart-rate alert	Backend API	Low	Result must be visible in mobile UI within expected time.	Pending QA
FRS-074-03	Support caregiver view	iOS	Not ranked	System shall complete the action without data loss.	In Review

FRS Section 75 - Security and privacy

This page belongs to Functional Requirements Specification for the Heart Rate Monitor mobile application. The application supports Android and iOS users, paired heart-rate sensors, manual BPM entry, trend review, alerts, data export, and controlled access. This block intentionally includes uneven numbering, repeated labels, partial owner fields, and mixed wording so an AI traceability/RAG engine can be tested against non-perfect documents. Page marker: FRS-075. Owner: Mobile Lead. Important terms: BPM, sensor pairing, low alert, high alert, caregiver, audit trail, local cache, encrypted sync, consent, export, notification.

- Unnumbered note - check sensor disconnect behavior during active monitoring.
- For iOS only? For Android only? unclear; parser should capture platform if visible.

63. FRS-075-3: App shall process heart rate data and preserve timestamp/timezone context.

Unnumbered note - check sensor disconnect behavior during active monitoring.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
FRS-075-01	Notification retry and suppression	Backend API	TBD	User shall receive clear error message on failure.	Approved?
FRS-075-02	Export readings as PDF/CSV	Android	TBD	Result must be visible in mobile UI within expected time.	In Review
FRS-075-03	Sync readings to cloud API	Android/iOS	Not ranked	TBD by clinical SME - intentionally incomplete.	Approved?
FRS_75.4	Display current BPM with timestamp	Local DB	Medium	TBD by clinical SME - intentionally incomplete.	Pending QA

FRS Section 76 - Exception handling

This page belongs to Functional Requirements Specification for the Heart Rate Monitor mobile application. The application supports Android and iOS users, paired heart-rate sensors, manual BPM entry, trend review, alerts, data export, and controlled access. This block intentionally includes uneven numbering, repeated labels, partial owner fields, and mixed wording so an AI traceability/RAG engine can be tested against non-perfect documents. Page marker: FRS-076. Owner: Clinical SME. Important terms: BPM, sensor pairing, low alert, high alert, caregiver, audit trail, local cache, encrypted sync, consent, export, notification.

64. Old requirement copied from web portal project - review if still valid.

Old requirement copied from web portal project - review if still valid.

65. Old requirement copied from web portal project - review if still valid.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
FRS-076-01	Capture heart rate from paired device	Android	Not ranked	TBD by clinical SME - intentionally incomplete.	Pending QA
FRS-076-02	Notification retry and suppression	Bluetooth Sensor	Medium	Audit/event record shall be generated where applicable.	Draft
FRS-076-03	Biometric login option	Backend API	Medium	Data shall be encrypted in transit and at rest.	Approved?

FRS Section 77 - API behavior

This page belongs to Functional Requirements Specification for the Heart Rate Monitor mobile application. The application supports Android and iOS users, paired heart-rate sensors, manual BPM entry, trend review, alerts, data export, and controlled access. This block intentionally includes uneven numbering, repeated labels, partial owner fields, and mixed wording so an AI traceability/RAG engine can be tested against non-perfect documents. Page marker: FRS-077. Owner: Product. Important terms: BPM, sensor pairing, low alert, high alert, caregiver, audit trail, local cache, encrypted sync, consent, export, notification.

- FRS-077-1: App shall process heart rate data and preserve timestamp/timezone context.

66. Acceptance: expected result to be confirmed by QA; evidence screenshot may be attached later.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
FRS_77.1	Manual heart rate entry when device unavailable	Backend API	Low	TBD by clinical SME - intentionally incomplete.	Pending QA
FRS-077-02	Capture heart rate from paired device	Cloud Sync	High	Audit/event record shall be generated where applicable.	Rejected once
FRS-077-03	Notification retry and suppression	Android/iOS	Low	TBD by clinical SME - intentionally incomplete.	In Review
FRS-077-04	Battery and Bluetooth warning	Backend API	Medium	User shall receive clear error message on failure.	Need SME

FRS Section 78 - Exception handling

missing owner

This page belongs to Functional Requirements Specification for the Heart Rate Monitor mobile application. The application supports Android and iOS users, paired heart-rate sensors, manual BPM entry, trend review, alerts, data export, and controlled access. This block intentionally includes uneven numbering, repeated labels, partial owner fields, and mixed wording so an AI traceability/RAG engine can be tested against non-perfect documents. Page marker: FRS-078. Owner: Product. Important terms: BPM, sensor pairing, low alert, high alert, caregiver, audit trail, local cache, encrypted sync, consent, export, notification.

For iOS only? For Android only? unclear; parser should capture platform if visible.

FRS-078-2: App shall process heart rate data and preserve timestamp/timezone context.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
FRS-078-01	Audit user actions	Notification Service	High	User shall receive clear error message on failure.	Rejected once
FRS-078-02	Audit user actions	Local DB	Not ranked	Audit/event record shall be generated where applicable.	Need SME
FRS-078-03	Show trend chart for 7/30/90 days	Android	Not ranked	Data shall be encrypted in transit and at rest.	Need SME
FRS_78.4	Store readings securely on device	Cloud Sync	Medium	Data shall be encrypted in transit and at rest.	Need SME

Random inserted note: duplicate title appears again below.

FRS Section 78 - Exception handling

FRS Section 79 - Traceability notes

This page belongs to Functional Requirements Specification for the Heart Rate Monitor mobile application. The application supports Android and iOS users, paired heart-rate sensors, manual BPM entry, trend review, alerts, data export, and controlled access. This block intentionally includes uneven numbering, repeated labels, partial owner fields, and mixed wording so an AI traceability/RAG engine can be tested against non-perfect documents. Page marker: FRS-079. Owner: TBD. Important terms: BPM, sensor pairing, low alert, high alert, caregiver, audit trail, local cache, encrypted sync, consent, export, notification.

Acceptance: expected result to be confirmed by QA; evidence screenshot may be attached later.

Old requirement copied from web portal project - review if still valid.

- For iOS only? For Android only? unclear; parser should capture platform if visible.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
FRS-079-01	Validate sensor data range	iOS	High	System shall complete the action without data loss.	In Review
FRS-079-02	Validate sensor data range	Android	Medium	Data shall be encrypted in transit and at rest.	Approved?
FRS-079-03	Validate sensor data range	iOS	High	System shall complete the action without data loss.	Approved?
FRS-079-04	Notification retry and suppression	Local DB	Not ranked	TBD by clinical SME - intentionally incomplete.	In Review

FRS Section 80 - Traceability notes

This page belongs to Functional Requirements Specification for the Heart Rate Monitor mobile application. The application supports Android and iOS users, paired heart-rate sensors, manual BPM entry, trend review, alerts, data export, and controlled access. This block intentionally includes uneven numbering, repeated labels, partial owner fields, and mixed wording so an AI traceability/RAG engine can be tested against non-perfect documents. Page marker: FRS-080. Owner: QA. Important terms: BPM, sensor pairing, low alert, high alert, caregiver, audit trail, local cache, encrypted sync, consent, export, notification.

- 67. Old requirement copied from web portal project - review if still valid.
- 68. Old requirement copied from web portal project - review if still valid.
- 69. Unnumbered note - check sensor disconnect behavior during active monitoring.

For iOS only? For Android only? unclear; parser should capture platform if visible.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
FRS-080-01	Audit user actions	Backend API	Not ranked	Result must be visible in mobile UI within expected time.	Approved?
FRS-080-02	Validate sensor data range	Local DB	TBD	Data shall be encrypted in transit and at rest.	Need SME
FRS-080-03	Handle offline mode	Local DB	Low	System shall complete the action without data loss.	Approved?
FRS-080-04	Store readings securely on device	Android/iOS	TBD	Result must be visible in mobile UI within expected time.	In Review

FRS Section 81 - Risk and control mapping

RAG TEST CHUNK

This page belongs to Functional Requirements Specification for the Heart Rate Monitor mobile application. The application supports Android and iOS users, paired heart-rate sensors, manual BPM entry, trend review, alerts, data export, and controlled access. This block intentionally includes uneven numbering, repeated labels, partial owner fields, and mixed wording so an AI traceability/RAG engine can be tested against non-perfect documents. Page marker: FRS-081. Owner: Product. Important terms: BPM, sensor pairing, low alert, high alert, caregiver, audit trail, local cache, encrypted sync, consent, export, notification.

70. For iOS only? For Android only? unclear; parser should capture platform if visible.
- Acceptance: expected result to be confirmed by QA; evidence screenshot may be attached later.
71. Old requirement copied from web portal project - review if still valid.
72. For iOS only? For Android only? unclear; parser should capture platform if visible.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
FRS-081-01	Validate sensor data range	Local DB	Medium	User shall receive clear error message on failure.	Approved?
FRS_81.2	Display current BPM with timestamp	Local DB	Medium	User shall receive clear error message on failure.	Draft
FRS-081-03	Store readings securely on device	Notification Service	Medium	Data shall be encrypted in transit and at rest.	Rejected once

MIXED TRACEABILITY LINE: URS-00X -> FRS?? -> DS component maybe -> IQ/OQ/PQ evidence TBD. This line is intentionally noisy.

FRS Section 82 - Alert workflow

This page belongs to Functional Requirements Specification for the Heart Rate Monitor mobile application. The application supports Android and iOS users, paired heart-rate sensors, manual BPM entry, trend review, alerts, data export, and controlled access. This block intentionally includes uneven numbering, repeated labels, partial owner fields, and mixed wording so an AI traceability/RAG engine can be tested against non-perfect documents. Page marker: FRS-082. Owner: Clinical SME. Important terms: BPM, sensor pairing, low alert, high alert, caregiver, audit trail, local cache, encrypted sync, consent, export, notification.

73. Unnumbered note - check sensor disconnect behavior during active monitoring.

- Unnumbered note - check sensor disconnect behavior during active monitoring.
- Old requirement copied from web portal project - review if still valid.
- Old requirement copied from web portal project - review if still valid.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
FRS-082-01	Store readings securely on device	iOS	TBD	Result must be visible in mobile UI within expected time.	Need SME
FRS-082-02	Consent capture and privacy notice	Cloud Sync	TBD	Data shall be encrypted in transit and at rest.	Draft
FRS-082-03	Display current BPM with timestamp	Local DB	High	Result must be visible in mobile UI within expected time.	Draft

FRS Section 83 - Evidence placeholder

This page belongs to Functional Requirements Specification for the Heart Rate Monitor mobile application. The application supports Android and iOS users, paired heart-rate sensors, manual BPM entry, trend review, alerts, data export, and controlled access. This block intentionally includes uneven numbering, repeated labels, partial owner fields, and mixed wording so an AI traceability/RAG engine can be tested against non-perfect documents. Page marker: FRS-083. Owner: TBD. Important terms: BPM, sensor pairing, low alert, high alert, caregiver, audit trail, local cache, encrypted sync, consent, export, notification.

For iOS only? For Android only? unclear; parser should capture platform if visible.

- Old requirement copied from web portal project - review if still valid.

Unnumbered note - check sensor disconnect behavior during active monitoring.

74. Acceptance: expected result to be confirmed by QA; evidence screenshot may be attached later.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
FRS-083-01	Sync readings to cloud API	Local DB	High	User shall receive clear error message on failure.	Need SME
FRS-083-02	Audit user actions	Android/iOS	High	Data shall be encrypted in transit and at rest.	Draft
FRS-083-03	Sync readings to cloud API	iOS	Medium	System shall complete the action without data loss.	In Review
FRS-083-04	Store readings securely on device	Cloud Sync	High	Result must be visible in mobile UI within expected time.	Draft

FRS Section 84 - Requirement details

This page belongs to Functional Requirements Specification for the Heart Rate Monitor mobile application. The application supports Android and iOS users, paired heart-rate sensors, manual BPM entry, trend review, alerts, data export, and controlled access. This block intentionally includes uneven numbering, repeated labels, partial owner fields, and mixed wording so an AI traceability/RAG engine can be tested against non-perfect documents. Page marker: FRS-084. Owner: Product. Important terms: BPM, sensor pairing, low alert, high alert, caregiver, audit trail, local cache, encrypted sync, consent, export, notification.

- 75. Unnumbered note - check sensor disconnect behavior during active monitoring.
- FRS-084-2: App shall process heart rate data and preserve timestamp/timezone context.
- 76. Old requirement copied from web portal project - review if still valid.
- 77. Old requirement copied from web portal project - review if still valid.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
FRS-084-01	Store readings securely on device	Notification Service	High	TBD by clinical SME - intentionally incomplete.	Need SME
FRS_84.2	Battery and Bluetooth warning	Local DB	Not ranked	Result must be visible in mobile UI within expected time.	Need SME
FRS-084-03	Battery and Bluetooth warning	Android/iOS	Not ranked	TBD by clinical SME - intentionally incomplete.	Approved?
FRS-084-04	Handle offline mode	Local DB	Medium	Audit/event record shall be generated where applicable.	Approved?

FRS Section 85 - API behavior

This page belongs to Functional Requirements Specification for the Heart Rate Monitor mobile application. The application supports Android and iOS users, paired heart-rate sensors, manual BPM entry, trend review, alerts, data export, and controlled access. This block intentionally includes uneven numbering, repeated labels, partial owner fields, and mixed wording so an AI traceability/RAG engine can be tested against non-perfect documents. Page marker: FRS-085. Owner: QA. Important terms: BPM, sensor pairing, low alert, high alert, caregiver, audit trail, local cache, encrypted sync, consent, export, notification.

- For iOS only? For Android only? unclear; parser should capture platform if visible.
- For iOS only? For Android only? unclear; parser should capture platform if visible.
- Old requirement copied from web portal project - review if still valid.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
FRS-085-01	Notification retry and suppression	Bluetooth Sensor	Not ranked	TBD by clinical SME - intentionally incomplete.	Approved?
FRS-085-02	Sync readings to cloud API	Notification Service	Low	Data shall be encrypted in transit and at rest.	Pending QA
FRS_85.3	Audit user actions	Backend API	Medium	Data shall be encrypted in transit and at rest.	Rejected once
FRS-085-04	Store readings securely on device	Cloud Sync	Not ranked	TBD by clinical SME - intentionally incomplete.	Need SME

FRS Section 86 - Evidence placeholder

This page belongs to Functional Requirements Specification for the Heart Rate Monitor mobile application. The application supports Android and iOS users, paired heart-rate sensors, manual BPM entry, trend review, alerts, data export, and controlled access. This block intentionally includes uneven numbering, repeated labels, partial owner fields, and mixed wording so an AI traceability/RAG engine can be tested against non-perfect documents. Page marker: FRS-086. Owner: QA. Important terms: BPM, sensor pairing, low alert, high alert, caregiver, audit trail, local cache, encrypted sync, consent, export, notification.

FRS-086-1: App shall process heart rate data and preserve timestamp/timezone context.

78. Acceptance: expected result to be confirmed by QA; evidence screenshot may be attached later.

- FRS-086-3: App shall process heart rate data and preserve timestamp/timezone context.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
FRS-086-01	Support caregiver view	Bluetooth Sensor	High	Result must be visible in mobile UI within expected time.	Draft
FRS_86.2	Export readings as PDF/CSV	Android	Not ranked	Audit/event record shall be generated where applicable.	Approved?
FRS-086-03	Generate high/low heart-rate alert	Bluetooth Sensor	High	TBD by clinical SME - intentionally incomplete.	Need SME
FRS-086-04	Support caregiver view	Local DB	Medium	Data shall be encrypted in transit and at rest.	Approved?
FRS-086-05	Consent capture and privacy notice	Local DB	Medium	System shall complete the action without data loss.	Draft

FRS Section 87 - Traceability notes

old text retained

This page belongs to Functional Requirements Specification for the Heart Rate Monitor mobile application. The application supports Android and iOS users, paired heart-rate sensors, manual BPM entry, trend review, alerts, data export, and controlled access. This block intentionally includes uneven numbering, repeated labels, partial owner fields, and mixed wording so an AI traceability/RAG engine can be tested against non-perfect documents. Page marker: FRS-087. Owner: Mobile Lead. Important terms: BPM, sensor pairing, low alert, high alert, caregiver, audit trail, local cache, encrypted sync, consent, export, notification.

Unnumbered note - check sensor disconnect behavior during active monitoring.

Unnumbered note - check sensor disconnect behavior during active monitoring.

79. For iOS only? For Android only? unclear; parser should capture platform if visible.

- Acceptance: expected result to be confirmed by QA; evidence screenshot may be attached later.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
FRS-087-01	Notification retry and suppression	Android	Not ranked	TBD by clinical SME - intentionally incomplete.	In Review
FRS-087-02	Support caregiver view	Cloud Sync	Medium	Data shall be encrypted in transit and at rest.	Draft
FRS-087-03	Notification retry and suppression	Cloud Sync	Medium	System shall complete the action without data loss.	Rejected once
FRS-087-04	Audit user actions	iOS	High	Data shall be encrypted in transit and at rest.	Rejected once

FRS Section 88 - Mobile platform variation

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80. Unnumbered note - check sensor disconnect behavior during active monitoring.

- Old requirement copied from web portal project - review if still valid.

For iOS only? For Android only? unclear; parser should capture platform if visible.

81. FRS-088-4: App shall process heart rate data and preserve timestamp/timezone context.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
FRS-088-01	Generate high/low heart-rate alert	Cloud Sync	TBD	TBD by clinical SME - intentionally incomplete.	Approved?
FRS-088-02	Generate high/low heart-rate alert	Notification Service	High	User shall receive clear error message on failure.	Need SME
FRS_88.3	Notification retry and suppression	Local DB	High	TBD by clinical SME - intentionally incomplete.	Draft

FRS Section 89 - Data handling notes

OCR noisy line

This page belongs to Functional Requirements Specification for the Heart Rate Monitor mobile application. The application supports Android and iOS users, paired heart-rate sensors, manual BPM entry, trend review, alerts, data export, and controlled access. This block intentionally includes uneven numbering, repeated labels, partial owner fields, and mixed wording so an AI traceability/RAG engine can be tested against non-perfect documents. Page marker: FRS-089. Owner: QA. Important terms: BPM, sensor pairing, low alert, high alert, caregiver, audit trail, local cache, encrypted sync, consent, export, notification.

82. Unnumbered note - check sensor disconnect behavior during active monitoring.

- Old requirement copied from web portal project - review if still valid.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
FRS-089-01	Capture heart rate from paired device	iOS	Not ranked	TBD by clinical SME - intentionally incomplete.	Rejected once
FRS-089-02	Sync readings to cloud API	Notification Service	Medium	TBD by clinical SME - intentionally incomplete.	Need SME
FRS-089-03	Audit user actions	Backend API	Not ranked	System shall complete the action without data loss.	Pending QA
FRS-089-04	Audit user actions	Android	Medium	System shall complete the action without data loss.	In Review
FRS-089-05	Export readings as PDF/CSV	Backend API	Medium	System shall complete the action without data loss.	Need SME

FRS Section 90 - Security and privacy

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- Unnumbered note - check sensor disconnect behavior during active monitoring.

83. Old requirement copied from web portal project - review if still valid.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
FRS-090-01	Battery and Bluetooth warning	Cloud Sync	Not ranked	User shall receive clear error message on failure.	Pending QA
FRS-090-02	Battery and Bluetooth warning	iOS	Not ranked	Audit/event record shall be generated where applicable.	In Review
FRS-090-03	Capture heart rate from paired device	Cloud Sync	Medium	Data shall be encrypted in transit and at rest.	In Review
FRS_90.4	Capture heart rate from paired device	Android/iOS	High	TBD by clinical SME - intentionally incomplete.	Need SME

MIXED TRACEABILITY LINE: URS-00X -> FRS?? -> DS component maybe -> IQ/OQ/PQ evidence TBD. This line is intentionally noisy.

FRS Section 91 - Mobile platform variation

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84. For iOS only? For Android only? unclear; parser should capture platform if visible.

- Acceptance: expected result to be confirmed by QA; evidence screenshot may be attached later.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
FRS-091-01	Audit user actions	Notification Service	High	Audit/event record shall be generated where applicable.	Need SME
FRS-091-02	Handle offline mode	Local DB	Medium	Audit/event record shall be generated where applicable.	Draft
FRS-091-03	Handle offline mode	Cloud Sync	TBD	Data shall be encrypted in transit and at rest.	Approved?
FRS-091-04	Export readings as PDF/CSV	iOS	High	System shall complete the action without data loss.	Pending QA
FRS-091-05	Store readings securely on device	Bluetooth Sensor	High	Audit/event record shall be generated where applicable.	Draft

Random inserted note: duplicate title appears again below.

FRS Section 91 - Mobile platform variation

FRS Section 92 - Audit trail mapping

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Acceptance: expected result to be confirmed by QA; evidence screenshot may be attached later.

- Old requirement copied from web portal project - review if still valid.

85. Acceptance: expected result to be confirmed by QA; evidence screenshot may be attached later.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
FRS-092-01	Show trend chart for 7/30/90 days	Android	High	User shall receive clear error message on failure.	In Review
FRS-092-02	Support caregiver view	Android/iOS	TBD	Result must be visible in mobile UI within expected time.	Draft
FRS-092-03	Consent capture and privacy notice	Android	Not ranked	System shall complete the action without data loss.	Draft
FRS-092-04	Manual heart rate entry when device unavailable	Cloud Sync	TBD	System shall complete the action without data loss.	Rejected once

FRS Section 93 - API behavior

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Acceptance: expected result to be confirmed by QA; evidence screenshot may be attached later.

- Old requirement copied from web portal project - review if still valid.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
FRS-093-01	Notification retry and suppression	Android/iOS	Low	System shall complete the action without data loss.	Need SME
FRS-093-02	Export readings as PDF/CSV	Android	Low	Result must be visible in mobile UI within expected time.	Rejected once
FRS-093-03	Sync readings to cloud API	Cloud Sync	High	User shall receive clear error message on failure.	Draft

FRS Section 94 - Traceability notes

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- FRS-094-1: App shall process heart rate data and preserve timestamp/timezone context.

86. Old requirement copied from web portal project - review if still valid.

For iOS only? For Android only? unclear; parser should capture platform if visible.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
FRS-094-01	Consent capture and privacy notice	Backend API	Medium	System shall complete the action without data loss.	Approved?
FRS-094-02	Support caregiver view	Backend API	Not ranked	Result must be visible in mobile UI within expected time.	Draft
FRS-094-03	Capture heart rate from paired device	Local DB	TBD	Audit/event record shall be generated where applicable.	Rejected once
FRS-094-04	Generate high/low heart-rate alert	Android/iOS	TBD	TBD by clinical SME - intentionally incomplete.	Approved?

FRS Section 95 - Evidence placeholder

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FRS-095-1: App shall process heart rate data and preserve timestamp/timezone context.

Unnumbered note - check sensor disconnect behavior during active monitoring.

- FRS-095-3: App shall process heart rate data and preserve timestamp/timezone context.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
FRS-095-01	Notification retry and suppression	Local DB	High	TBD by clinical SME - intentionally incomplete.	Pending QA
FRS-095-02	Validate sensor data range	Local DB	Medium	User shall receive clear error message on failure.	In Review
FRS_95.3	Export readings as PDF/CSV	Notification Service	Not ranked	Data shall be encrypted in transit and at rest.	Need SME

FRS Section 96 - Bluetooth edge cases

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87. FRS-096-1: App shall process heart rate data and preserve timestamp/timezone context.

Acceptance: expected result to be confirmed by QA; evidence screenshot may be attached later.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
FRS-096-01	Validate sensor data range	Backend API	Low	Audit/event record shall be generated where applicable.	Approved?
FRS_96.2	Export readings as PDF/CSV	Android	Not ranked	Audit/event record shall be generated where applicable.	Pending QA
FRS_96.3	Display current BPM with timestamp	Local DB	TBD	Audit/event record shall be generated where applicable.	Draft
FRS-096-04	Manual heart rate entry when device unavailable	iOS	High	Result must be visible in mobile UI within expected time.	Draft
FRS-096-05	Battery and Bluetooth warning	Backend API	High	Result must be visible in mobile UI within expected time.	Need SME

FRS Section 97 - Alert workflow

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- For iOS only? For Android only? unclear; parser should capture platform if visible.
88. Acceptance: expected result to be confirmed by QA; evidence screenshot may be attached later.
- FRS-097-3: App shall process heart rate data and preserve timestamp/timezone context.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
FRS-097-01	Show trend chart for 7/30/90 days	Backend API	Low	Data shall be encrypted in transit and at rest.	Pending QA
FRS_97.2	Consent capture and privacy notice	Backend API	TBD	System shall complete the action without data loss.	Rejected once
FRS-097-03	Store readings securely on device	iOS	Not ranked	Result must be visible in mobile UI within expected time.	Draft

FRS Section 98 - API behavior

This page belongs to Functional Requirements Specification for the Heart Rate Monitor mobile application. The application supports Android and iOS users, paired heart-rate sensors, manual BPM entry, trend review, alerts, data export, and controlled access. This block intentionally includes uneven numbering, repeated labels, partial owner fields, and mixed wording so an AI traceability/RAG engine can be tested against non-perfect documents. Page marker: FRS-098. Owner: Validation. Important terms: BPM, sensor pairing, low alert, high alert, caregiver, audit trail, local cache, encrypted sync, consent, export, notification.

- Acceptance: expected result to be confirmed by QA; evidence screenshot may be attached later.

FRS-098-2: App shall process heart rate data and preserve timestamp/timezone context.

89. Unnumbered note - check sensor disconnect behavior during active monitoring.
90. FRS-098-4: App shall process heart rate data and preserve timestamp/timezone context.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
FRS-098-01	Capture heart rate from paired device	iOS	High	User shall receive clear error message on failure.	Approved?
FRS-098-02	Generate high/low heart-rate alert	Android/iOS	Low	Result must be visible in mobile UI within expected time.	Need SME
FRS-098-03	Battery and Bluetooth warning	Notification Service	High	Audit/event record shall be generated where applicable.	Rejected once

FRS Section 99 - Security and privacy

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91. Old requirement copied from web portal project - review if still valid.

Unnumbered note - check sensor disconnect behavior during active monitoring.

FRS-099-3: App shall process heart rate data and preserve timestamp/timezone context.

- For iOS only? For Android only? unclear; parser should capture platform if visible.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
FRS_99.1	Capture heart rate from paired device	iOS	TBD	System shall complete the action without data loss.	Rejected once
FRS-099-02	Show trend chart for 7/30/90 days	Bluetooth Sensor	TBD	Data shall be encrypted in transit and at rest.	Draft
FRS-099-03	Handle offline mode	iOS	High	Audit/event record shall be generated where applicable.	Approved?
FRS-099-04	Generate high/low heart-rate alert	Android/iOS	TBD	Data shall be encrypted in transit and at rest.	Need SME
FRS-099-05	Generate high/low heart-rate alert	iOS	TBD	System shall complete the action without data loss.	Pending QA

MIXED TRACEABILITY LINE: URS-00X -> FRS?? -> DS component maybe -> IQ/OQ/PQ evidence TBD. This line is intentionally noisy.

FRS Section 100 - Alert workflow

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92. FRS-100-1: App shall process heart rate data and preserve timestamp/timezone context.

- FRS-100-2: App shall process heart rate data and preserve timestamp/timezone context.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
FRS-100-01	Support caregiver view	Android/iOS	Medium	System shall complete the action without data loss.	Need SME
FRS-100-02	Validate sensor data range	iOS	Low	User shall receive clear error message on failure.	Approved?
FRS-100-03	Store readings securely on device	Android/iOS	High	Data shall be encrypted in transit and at rest.	Rejected once
FRS-100-04	Show trend chart for 7/30/90 days	Notification Service	TBD	Data shall be encrypted in transit and at rest.	Draft